

A QUESTION TO OPEN WITH

Someone designed every neural  
network you have ever used.

*What if no one had to?*

# Scaling Adaptive Substrate Neuroevolution

*Diagnosing why evolution-designed networks never scaled,  
then building the platform that lets them.*

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University of Neuchâtel, Switzerland

11 August 2026

# I set out to build one thing, and had to build something else first.

01 · QUESTION

2020

Meta-cognition and meta-learning. Can a machine learn how to learn, and question its own reasoning?

02 · ANSWER

GEENNS

An architecture that grows its own intelligence out of frozen, reusable parts.

03 · THE WALL

Infeasible

The one algorithm that discovers its own structure is sequential. A single modest GEENNS run costs years of CPU.

04 · FAILED FIX

Not enough

Four strategies to make the existing algorithm faster. Each bought a constant factor. The cost grows with depth.

05 · REBUILD

EMR

A reformulation, not an optimization. The gap widens with depth, and the goal becomes testable.

■ ACT I

# I

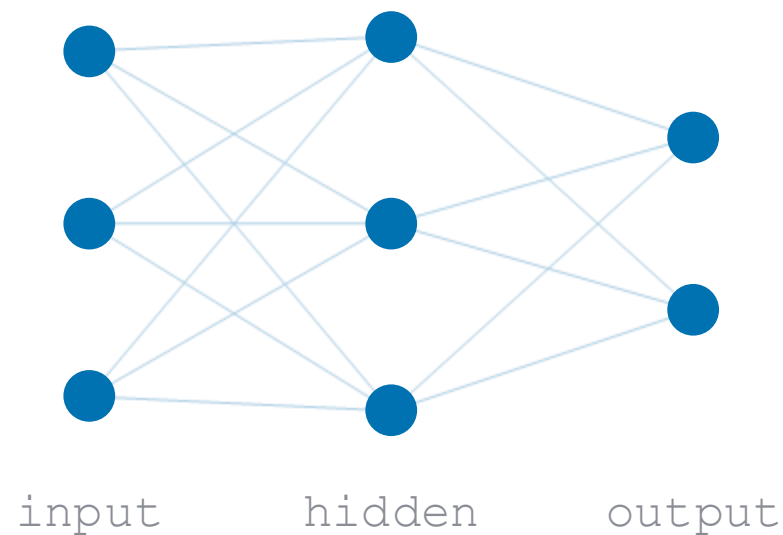
# The Premise

Why that goal needs evolution, and not better design.

# Design the network, or evolve it.

## DESIGNED

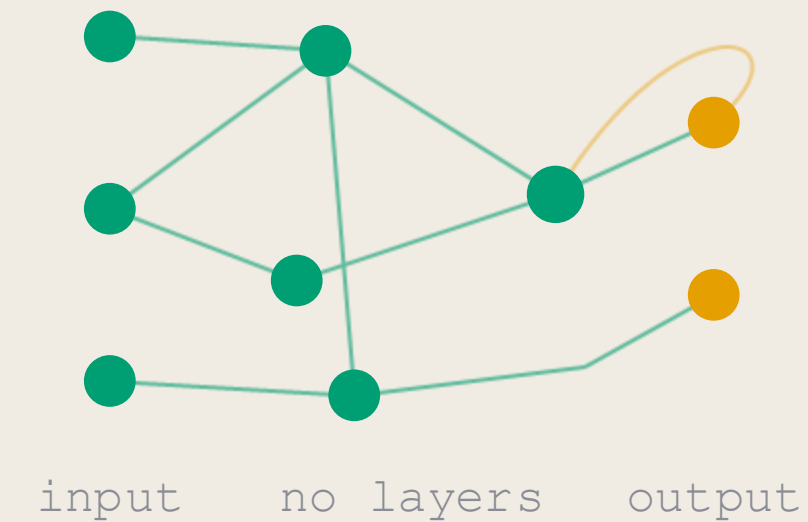
A human fixes the shape; the machine fills in the weights.



Topology frozen before training. Layers, connections, activations: every choice is human intuition, and it caps what the system can become.

## EVOLVED

Evolution discovers the shape and the weights together.



No template, no human bias. The result is **sparse by nature** : evolution pays only for connections it needs. Optimized for the task, not readability.

# DNA is a recipe, not a blueprint.

BIOLOGY

3<sup>B</sup>

base pairs in  
the genome

*grow*

10<sup>14</sup>

synapses in the  
brain

A description **five orders of magnitude** smaller than the thing it builds. No gene says *five fingers* : a chemical gradient does, and every hand gets five anyway.

COMPUTATION

one CPPN

the evolved  
genome, queried

*query*

the substrate

every neuron and  
weight

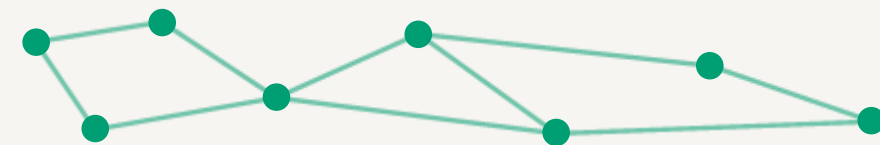
Indirect encoding rebuilds that same asymmetry. The difficulties in this thesis are **the price of working in this regime** , not arguments against it.

Early neuroevolution wrote every connection into the genome directly, and threw the asymmetry away. Structure is not stored. It emerges from running the rule.

# Each step hands more of the design to evolution.

2002

## NEAT

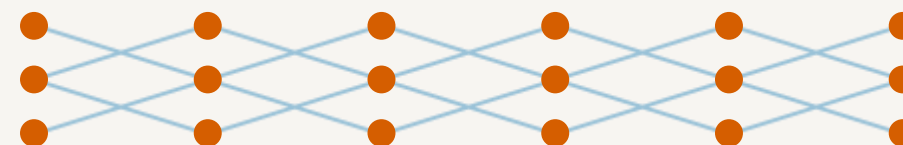


neurons **evolved** weights **evolved**  
geometry **none**

Two lists: **node genes** (id, type) and **connection genes** (in, out, weight, enable bit, innovation number). Purely topological, **no coordinates**, one gene per edge.

2009

## HyperNEAT



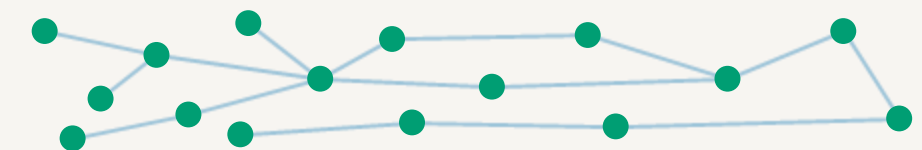
neurons **human** connections **CPPN**  
encoding **indirect**

A human lays the neurons on a **fixed grid**. A **CPPN** discovers every connection *between* those pre-placed nodes, compressing millions of weights into a tiny genome.

2010 · the starting point

iterated 2011 · journal 2012

## ES-HyperNEAT

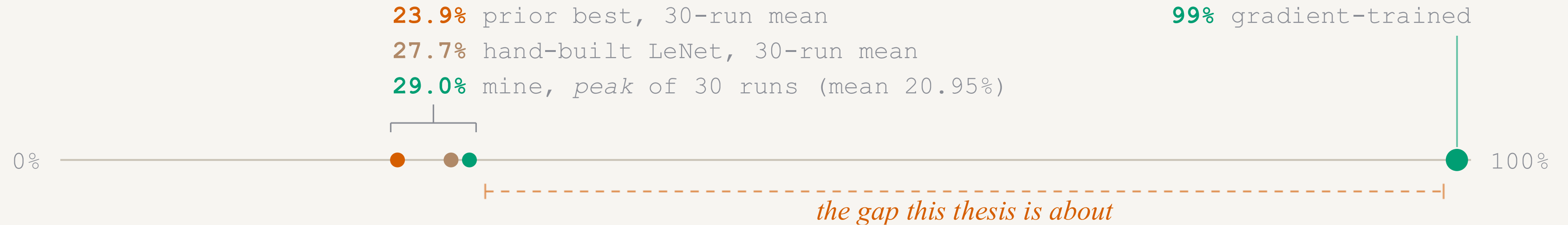


neurons **discovered** connections **CPPN** encoding **indirect**

The grid goes too. A **quadtree** discovers **where the neurons belong**, adding resolution wherever the CPPN encodes detail. The last human decision is gone.

Geometry enters in column two, **drawn by hand**. Column three evolves it, using a **quadtree**.

# The promise is real. The practical utility was not.



**320×** fewer evaluations **per run**,  $100 \times 20$  vs  $256 \times 2,500$

**self-found** a far smaller substrate, **discovered not drawn**

Every result sits in that cluster, mine included. Not a search failure to be tuned away, but **three structural barriers**, each needing a different solution.

## BARRIER 01

### Configuration

A search problem. The viable region is real but hidden.

billions of configurations

## BARRIER 02

### Representation

An architectural problem. The network sees almost nothing.

96% of the image unseen

## BARRIER 03

### Computation

An algorithmic problem. It cannot use parallel hardware.

no GPU, at any scale

■ ACT II

# II

## The Diagnosis

Three walls between the goal and a tool that could reach it. All three structural.

# Over three billion ways to be wrong.

$3 \times 10^9$

configurations in just a 16-  
parameter subspace MNIST  
classification

~40

interacting parameters

months

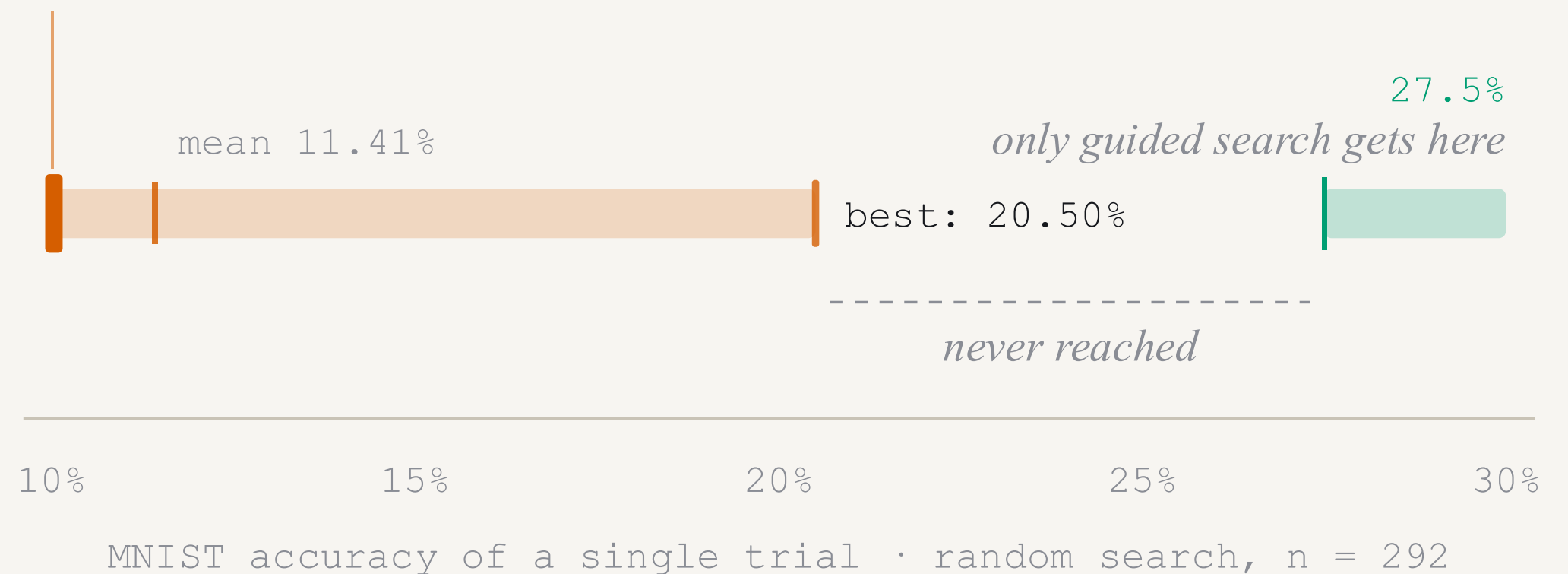
of compute for one sweep

$R^2 = .01$

TPE trend across 2,013 trials

MEDIAN = MINIMUM = 10.00%

*so at least half of all trials score  
exactly the random-guessing floor*



A search problem → and a **binary** one: there is no gradient to follow.

# Search smarter, then quit early.

## PAPER 1 · TPE OPTIMIZATION

Bayesian search finds the needle.

**29.0%** validated MNIST peak prior  
best, 30-run means: 23.9%  
generic, 27.7% LeNet

2,013 trials concentrated on the productive region,  
and confirmed the ceiling is **structural, not a  
search failure**.

## PAPER 2 · EARLY-STOPPING PRUNER

Predict failure at generation 3.

**-41.6%** **0.872**  
compute cost  $F_1$  of the classifier

Same fitness quality, 64% fewer generations than  
Hyperband. This is what made the rest of the thesis  
affordable.

## DISCOVERY

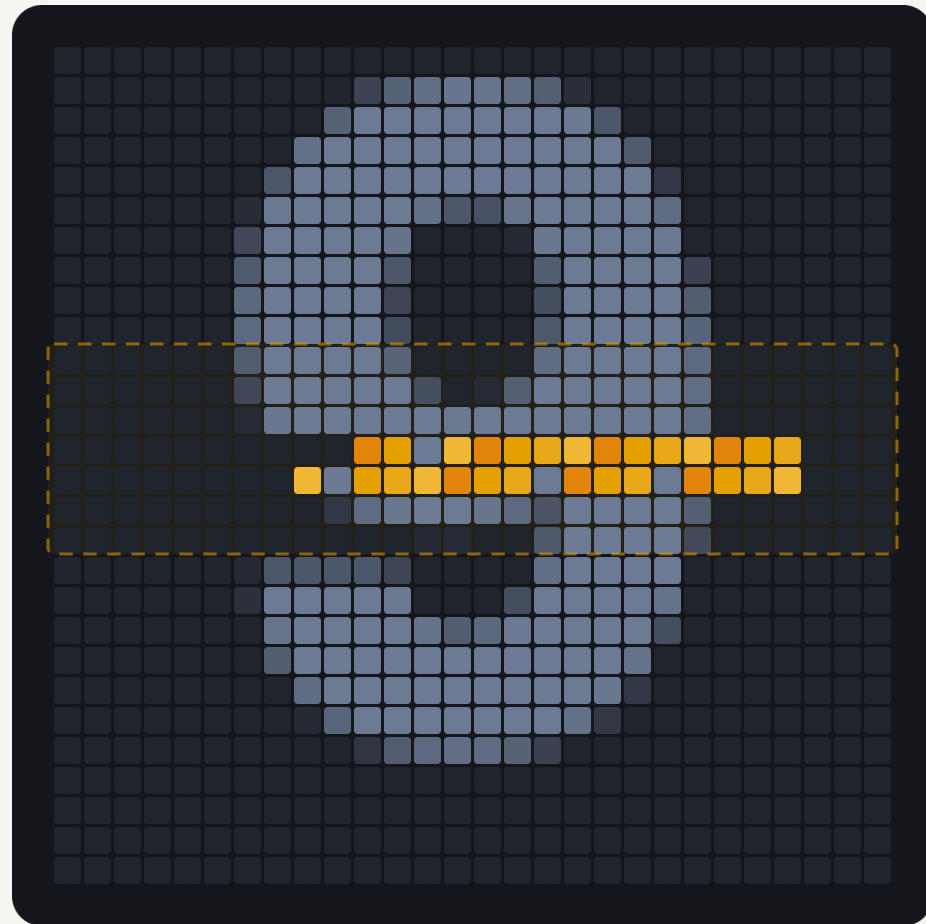
The **dynamics** are bimodal: winners and  
losers separate at generation 2, and by  
generation 3 the gap is  $\Delta = 0.037$ , enough  
to decide.

## WHY QUITTING EARLY PAYS

**93%** at the 10% floor, **generation 0**

**29%** **never leave it**, all 17 gens

**87%** stuck at gen 3 → **stuck all run**



one measured run dashed: rows 10–16, where  
all 30 runs fall

# It was looking at 3.6% of the picture.

**28** of 784 px reach the network. The other 96% is invisible to it.

The CPPN's building blocks all peak their variance near the vertical center, so the quadtree resolves along that midline and never leaves it. Across all 30 replications, **every active pixel fell inside rows 10–16** : a band covering a quarter of the image height.

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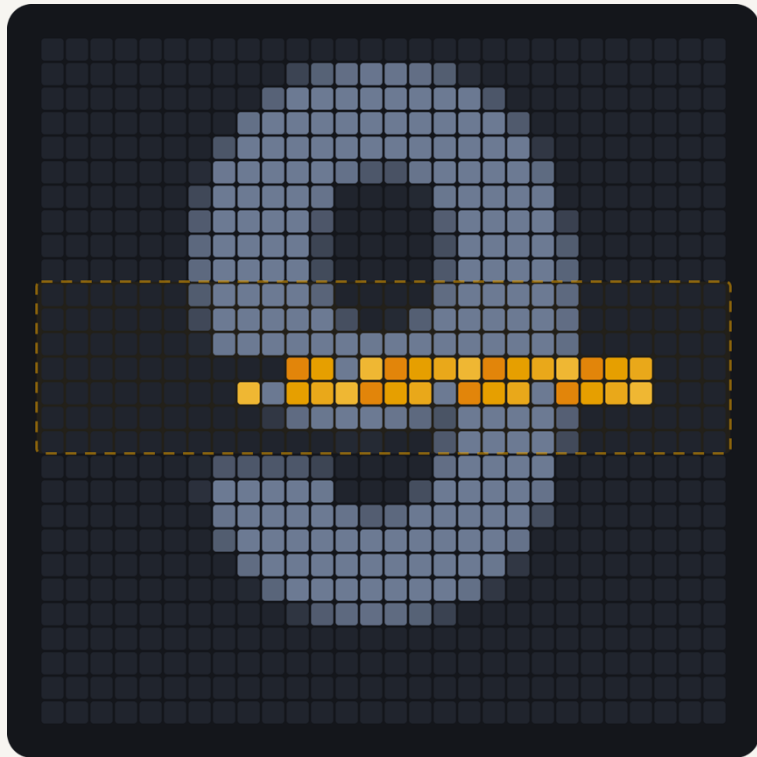
The network never sees the data. No amount of tuning fixes an information failure.

# Force it to look everywhere.

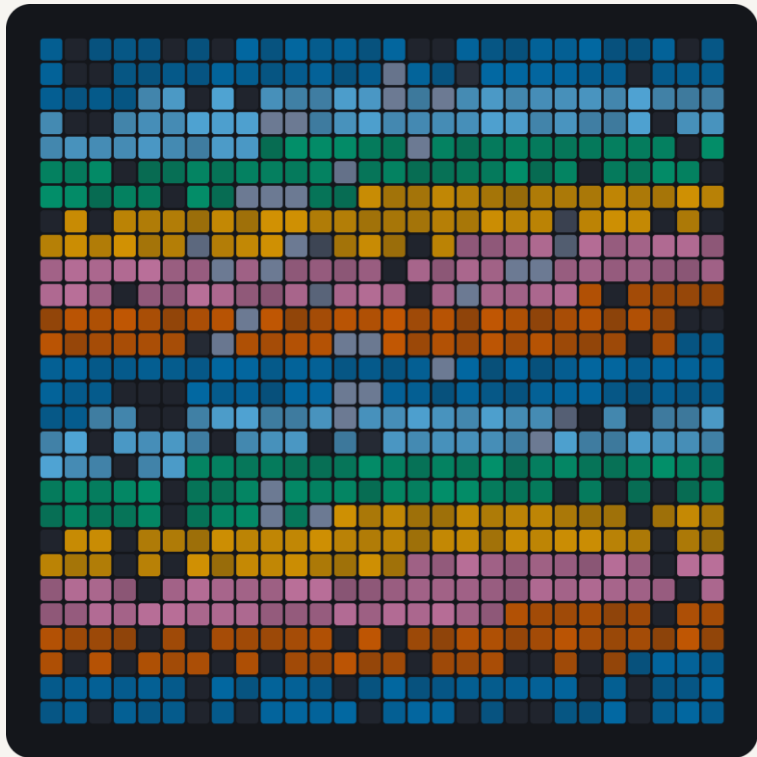
Split the 784-pixel input into **13 equal segments: one expert per band** . Each is forced to attend to its own slice, so the full input is finally represented.

| PIXEL COVERAGE · PEAK | MNIST ACCURACY · MEAN |
|-----------------------|-----------------------|
| 3.6% → 85%            | 21% → 43%             |

×2 accuracy · Cohen's d = 8.70 · peak run 50%  
first evidence for specialist composition



one substrate  
28 px in one run



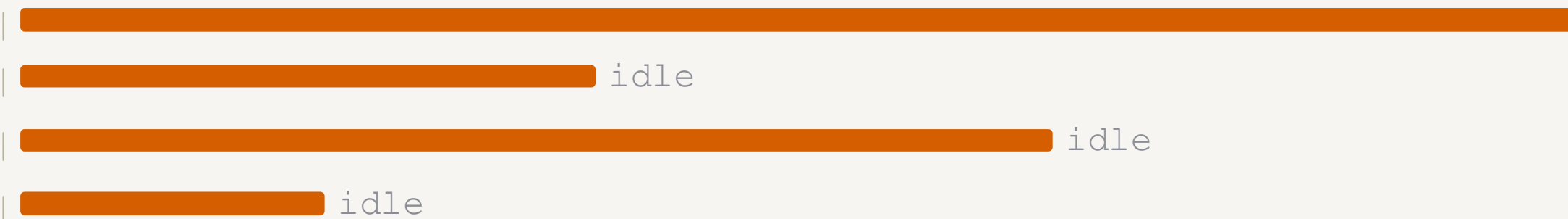
13 experts, one per band  
85%: the full input

# Every genome builds a different network, so the GPU can't help.

UNIFORM WORK → GPU FLIES



RAGGED QUADTREES → GPU STALLS



GPUs need every lane doing the **same work at once**.

The quadtree guarantees the opposite: a unique shape per genome.

0.54

$\eta^2$ : depth  
dominates  
runtime

1.65×

ceiling of 4  
tuning  
strategies  
combined

VERDICT

The quadtree must be **replaced, not optimized**. Until it is, one modest GEENNS run costs **~7 years of CPU**. (Paper 4, preprint.)

■ ACT III

# III

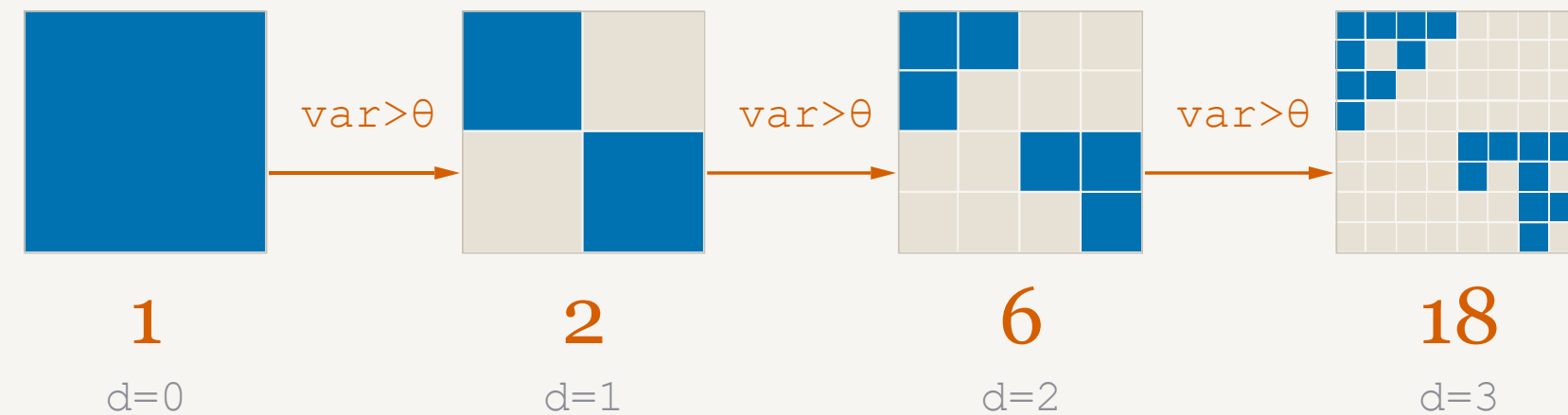
## The Breakthrough

The algorithm I had to build, and the single idea underneath it.

# Evaluate everything, filter later.

## ES-HyperNEAT · lazy

`subdivide_if(var >  $\theta$ )`



cells kept per level      `substrate = 18` (leaves only)

One tree per genome, no two alike, so nothing batches.

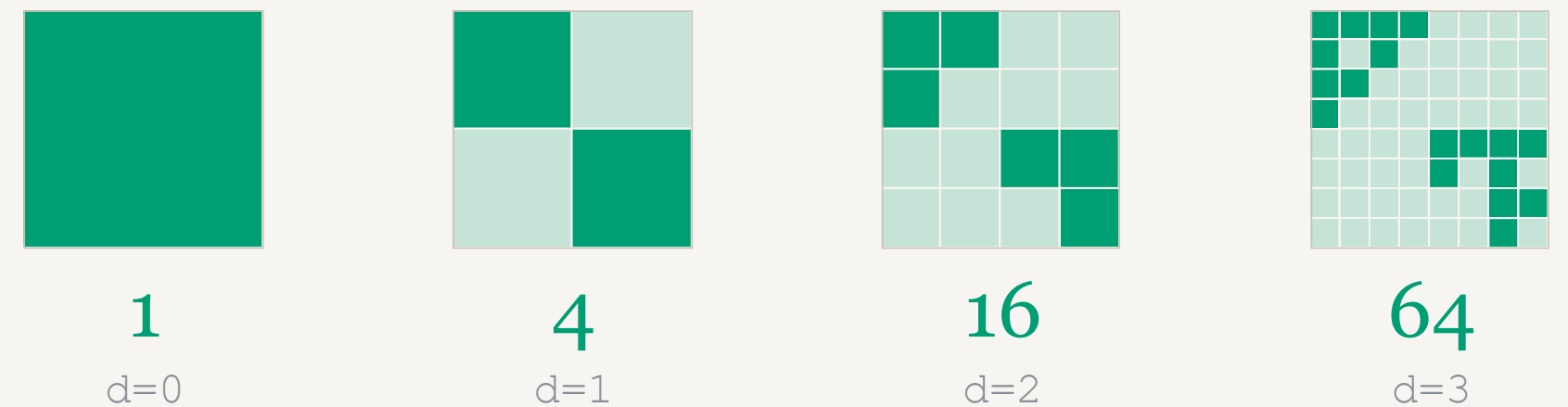
**Sequential.** Each level waits on the one above.

GUARANTEED CORRECT

EMR finds a **superset** of the quadtree's positions; identical sets give identical networks.

## EMR-HyperNEAT · eager

`eval_all(); filter(var >  $\theta$ )`



cells evaluated per level      `substrate = 27` (every level)

`one vmap call → variance → mask`

One grid, shared by the whole population and reused across problems.

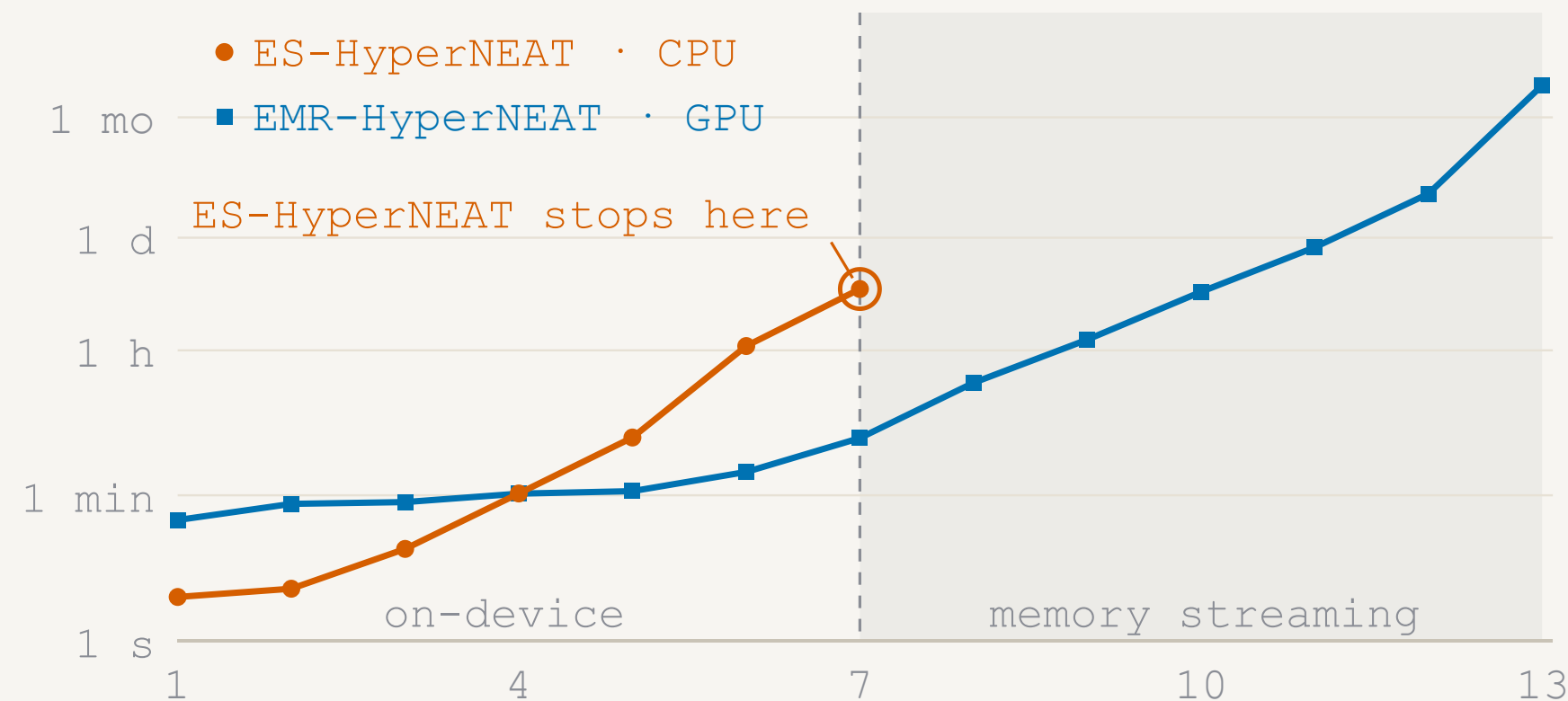
**Parallel.** Same cells, every level kept: a superset.

THE TRADE

We pay **redundant queries**, we buy **parallelism**. On small, short runs the quadtree still wins.

# Months of computation, done in days.

TOTAL RUNTIME vs DEPTH · XOR · 30 GENERATIONS Apple M4 Max CPU vs RTX 2080 Ti GPU



GEENNS PIPELINE · 34 CPPNs XOR SCALE

~250 CPU-days → 4.9 GPU-days

EMR alone: 8.3. The pruner takes it to 4.9. At MNIST scale the quadtree cost is on the order of **years**.

100%

solve, depths 4–7 (pop 1000; ES-HN: 0–33%)

15

problems across 4 domains validated

depth 13

reached, infeasible under the quadtree

any HW

CPU or GPU, same uniform tensors

# IV

## What It Unlocks

Three papers already published, and the goal from station two, now testable.

# The speedup was the least of it.

Every neuron is a row in a tensor, so **new biology is a new column** , not a new algorithm.

Each channel became a research direction that was impossible before EMR. Anatomy: B8.

## $f$ PER-NODE ACTIVATION

Every neuron picks from an 18-function palette.

**100%** parity, any bit-width,  
monotonic: 0%

A **categorical** divide, not a gradual one: oscillation solves it, monotonicity never does.

ALIFE 2026

## $\Sigma$ PALETTE EVOLUTION

Which functions belong in the palette, and when?

**2.1×** faster convergence 6  
biological families

Clonal selection, circadian, immune memory, **3,000 runs** .  
The last keeps its vocabulary across tasks: 88%.

PPSN 2026

## $r$ NEUROMODULATION

One substrate holding several behaviors at once.

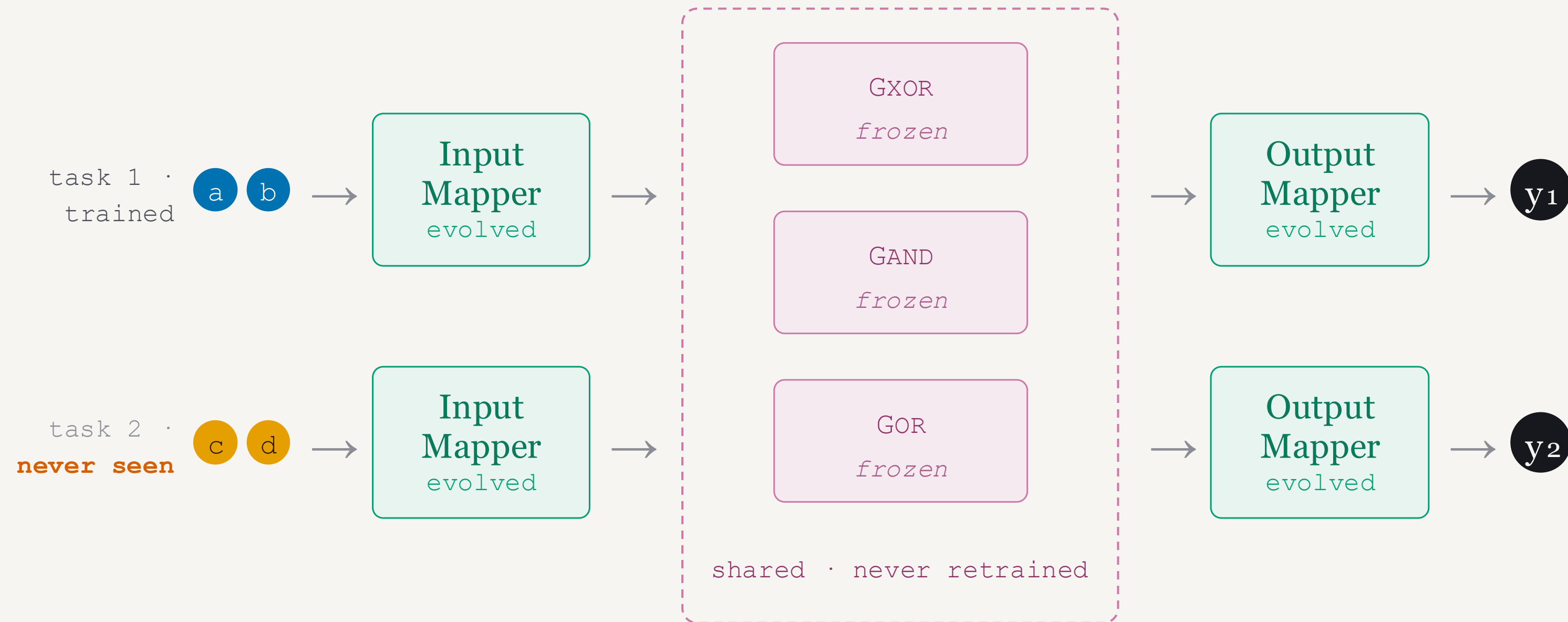
**5tasks** on one frozen topology,  
100% success

A per-task signal gates each neuron's receptors, so behavior switches **without touching a weight** .

ALIFE 2026

# A mind made of frozen, reusable parts.

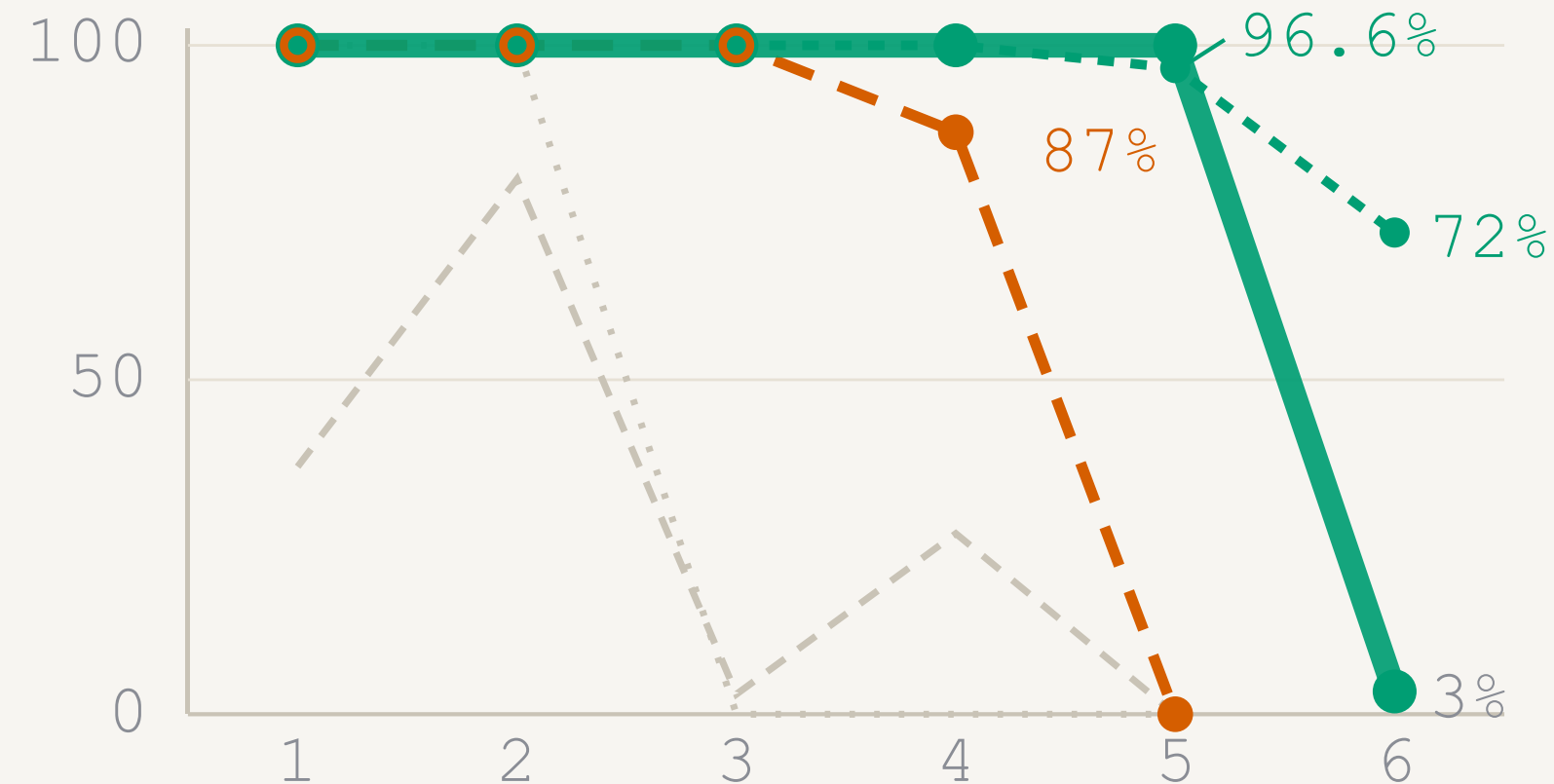
Evolve specialist grids, freeze them, and evolve small **mappers** that route inputs and compose outputs.  
Inspired by the one repeated microcircuit of the neocortical column.



A new task costs only **new mappers**, so knowledge accumulates instead of being overwritten.

# Frozen specialists solve what monoliths can't.

SUCCESS RATE vs GRIDS COMPOSED · 1-6 · N≈30 SEEDS



— composition of frozen grids: 100% to 5, then 3%

— strongest monolith (per-node f): 100 → 87 → 0

8× slower than composition at 4 grids

— composition + **gating**, evolve which grids to use

— uniform tanh / sin monoliths

At 5 grids **every** monolithic baseline collapses.

**57–93%**

on **new compound tasks** the grids never saw, no retraining. With gating: 100%.

**~3,100**

runs, >40 conditions

**$p < 1e-17$**

5-grid gap, Fisher

AND THE BAR WAS RAISED

≤ 4 grids floor low, standard bar (**0.95**) holds

5 grids 1,024 rows · floor 0.982 → bar **0.99**

6 grids 4,096 rows · floor 0.991 → bar **0.995**

Failures there plateau at **0.992–0.993** and get **every positive wrong**.

# What's still open, and why that's the exciting part.

## THE GAP

Individual mappers learn **task-specific routing**, not general compositional strategy. Zero-shot fitness lands at  $0.53-0.80$ , below the  $0.95$  bar.

And I assigned each grid its job by hand. So whether specialization would **emerge** on its own is untested, not answered.

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$p = .112$  and the 4-grid success gap is **not significant**: the speed inversion is the significant result

## THE TWIST

Test every individual of a population evolved for one task on an **unseen** composition. Some already solve it, **latent solvers**.

$0.3\%$  of the population, vs  $<0.01\%$  in random ones at  $10\times$  the budget

The same individuals solve both targets (Jaccard  $1.0$ ), so it is shared structure, but it lives across the **population**, not any one mapper.

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$45-351\times$  **warm-starting**, seeding a new search with a population evolved on a related task ( $45-79\times$  at 3 grids,  $223-351\times$  at 5)

## THE REFRAME

Not "*does it generalize?*" but "**how do we distill the population signal into one mapper?**", a question this thesis finally makes it possible to ask.

■ ACT V

V

# The Horizon

What all of this is really for: the study of artificial life.

# This was never about beating gradient descent.

| WHAT GRADIENTS DO                            | WHAT EVOLUTION DOES                                         |
|----------------------------------------------|-------------------------------------------------------------|
| the shape      given: a <b>human drew it</b> | the shape <b>discovered:</b> the whole point                |
| what it tunes    weights, continuously       | what it searches    topology: discrete, un-differentiable   |
| on MNIST <b>&gt;99%</b> , and it should be   | on MNIST <b>43%</b> , a diagnostic, not a score             |
| the cost      ~100× more neurons             | why no gradient    no gradient for a <i>choice of graph</i> |

Not rivals, **complementary**. Gradients optimize *within* a structure; evolution proposes the structure. The hybrid is future work.

The contribution is **diagnostic and infrastructural** : name the barriers, prove which are structural, break them, and reach the *scale where the real question becomes answerable* .

■ THE QUESTION UNDERNEATH THE WHOLE THESIS

Can evolution, rather than engineering, shape networks whose behaviors *imitate no human*, and that no human hand has designed?

The benchmarks were always proxies. The real target is **emergent behavior**, solutions arrived at in their own way. This thesis built the engine to grow it. What is still missing is an instrument to measure it that does not quietly measure *us*.

# The emergence of *Artificialkind.*

Not artificial humans. Imitation was always the wrong goal. Systems whose behaviors **emerge** from evolution rather than instruction, worth studying on their own terms: *how they solve, why that way, and what we can learn from it.*

Bio-inspiration as a route to emergence, not mimicry. This thesis made it studiable at scale. Next: measuring it, see B14/B15.

# What the thesis delivers.

| CONTRIBUTION             | WALL    | KEY RESULT                                                        | STATUS      |
|--------------------------|---------|-------------------------------------------------------------------|-------------|
| TPE optimization         | Config  | 29.0% MNIST across 2,013 trials · first systematic sweep          | ● published |
| Early-stopping pruner    | Config  | −41.6% cost, $F_1 = 0.872$ · revealed a bimodal fitness landscape | ● published |
| Central bias + experts   | Repr.   | 3.6%→85% peak coverage; 21%→43% accuracy (d = 8.70)               | ● published |
| Quadtree incompatibility | Compute | Depth dominates runtime ( $\eta^2=0.54$ ); 1.65× tuning ceiling   | 🕒 preprint  |
| EMR-HyperNEAT            | Compute | Lazy→eager tensor reformulation; 12–34× at D5–7 · the platform    | ● published |

~14,500

experimental runs

4 + 1

papers in the thesis (published + preprint)

+ 3

further papers built on this foundation

■ IN CLOSING

*"The best prophets lead you up to the curtain and let you peer through for yourself."*

— Frank Herbert

This thesis doesn't deliver artificial life. It walks up to the curtain, **fast enough, broad enough, flexible enough**, and lets us peer through. The foundation is built. [The question is finally askable.](#)

# Thank you.

*Questions and discussion very welcome.*

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With thanks to Paul Cotofrei · Kilian Stoffel · Michael  
O'Neill (UCD) Arthur Gygax · Michael Palma Mendes · the  
committee

Romain Claret  
University of Neuchâtel

11 August 2026

Name a topic, I can go straight to it.

| BACKUP · DETAIL |                         |    | QUESTIONS · CH 1–3 |                           |    | CH 7 · THE CHALLENGES |                          |    | CH 7 · THE MECHANISM    |                           |    |
|-----------------|-------------------------|----|--------------------|---------------------------|----|-----------------------|--------------------------|----|-------------------------|---------------------------|----|
| B1              | Evolution vs gradients  | 31 | Q1                 | Why the detour            | 46 | Q21                   | Latent solvers           | 66 | Q37                     | The prototype as built    | 82 |
| B2              | Statistical rigor       | 32 | Q2                 | One barrier or three?     | 47 | Q22                   | The 4-grid inversion     | 67 | Q38                     | How gating works          | 83 |
| B3              | The 43% ceiling         | 33 | Q3                 | Why exactly three?        | 48 | Q23                   | Freezing, CPPN growth    | 68 | Q39                     | Correct vs distractor     | 84 |
| B4              | Central-bias mechanism  | 34 | Q4                 | Never integrated?         | 49 | Q24                   | Boolean = intelligence?  | 69 | Q40                     | How gates evolve          | 85 |
| B5              | The quadtree, explained | 35 | Q5                 | Lineage, no baseline      | 50 | Q25                   | Is the mapper cheating?  | 70 | Q41                     | What a continuous gate is | 86 |
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| B7              | EMR internals           | 37 | Q7                 | Measured or projected?    | 52 | Q27                   | Is biology load-bearing? | 72 | Q43                     | Why population-level      | 88 |
| B8              | Substrate as a tensor   | 38 | Q8                 | Is it a barrier?          | 53 | Q28                   | Who chose the parts?     | 73 | Q44                     | What warm-start changes   | 89 |
| B9              | Generalization gap      | 39 | Q9                 | The TPE search            | 54 | Q29                   | Not peer-reviewed        | 74 | CH 8                    |                           |    |
| B10             | Related work            | 40 | Q10                | How the rule is derived   | 55 | Q30                   | Is 5 grids the ceiling?  | 75 | Q45 The weakest part 90 |                           |    |
| B11             | Reproducibility         | 41 | Q11                | How the rule is validated | 56 | Q31                   | Does reuse pay off?      | 76 |                         |                           |    |
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| B14             | Measuring emergence     | 44 | Q13                | Central-bias scope        | 58 | Q34                   | Budget-matched?          | 79 |                         |                           |    |
| B15             | The next program        | 45 | Q14                | Why 13 experts?           | 59 | Q35                   | The excluded seed        | 80 |                         |                           |    |
|                 |                         |    | Q15                | Coverage vs fitness       | 60 | Q36                   | What did it learn?       | 81 |                         |                           |    |
|                 |                         |    | Q16                | One configuration?        | 61 |                       |                          |    |                         |                           |    |
|                 |                         |    | Q17                | Eager vs lazy             | 62 |                       |                          |    |                         |                           |    |
|                 |                         |    | Q18                | The equivalence theorem   | 63 |                       |                          |    |                         |                           |    |
|                 |                         |    | Q19                | Beyond Booleans           | 64 |                       |                          |    |                         |                           |    |
|                 |                         |    | Q20                | Is the divide real?       | 65 |                       |                          |    |                         |                           |    |

# Evolution and gradients work at different levels.

## THREE REASONS THIS IS EVOLUTIONARY

### 1 · Discovery is discrete

Placing a node from output variance is non-differentiable, so there is no gradient to follow.

### 2 · Topology is the search space

Node counts and connectivity, not weights alone.  
Relaxations (DARTS) bring their own limits.

### 3 · Geometric regularity

CPPNs exploit symmetry & repetition that gradients on a weight matrix can't capture.

## AND NAS?

NAS searches *within* a human-designed space, where the building blocks and connectivity are hand-specified. It selects among known components; it cannot discover unknown ones.

## COMPLEMENTARY, NOT RIVAL

One explicit future direction is a hybrid: EMR discovers topology, gradients train the weights.

# Median-based, non-parametric, effect-size-first.

**N=30**

seeds per condition (community standard); GPU timing N=3, flagged as a limitation

**median**

central tendency, robust to the outliers in evolutionary fitness distributions

**95% CI**

bootstrap intervals, 10,000 resamples

Non-parametric tests

Mann-Whitney U (two groups);  
Kruskal-Wallis (multi-group);  $\alpha = 0.05$

Effect sizes always

Cohen's d · rank-biserial r · Cliff's  $\delta$  ·  $\eta^2$ , reported with every p-value

Equivalence via TOST

e.g. the pruner preserves quality: tested, not inferred from a null result

■ B3 THE 43% MNIST CEILING: IS IT USEFUL?

# The same number, in three fields.

## IN DEEP LEARNING

A failure. state of the art exceeds 99%

## IN NEUROEVOLUTION

A big advance. +105% over baseline, d = 8.70

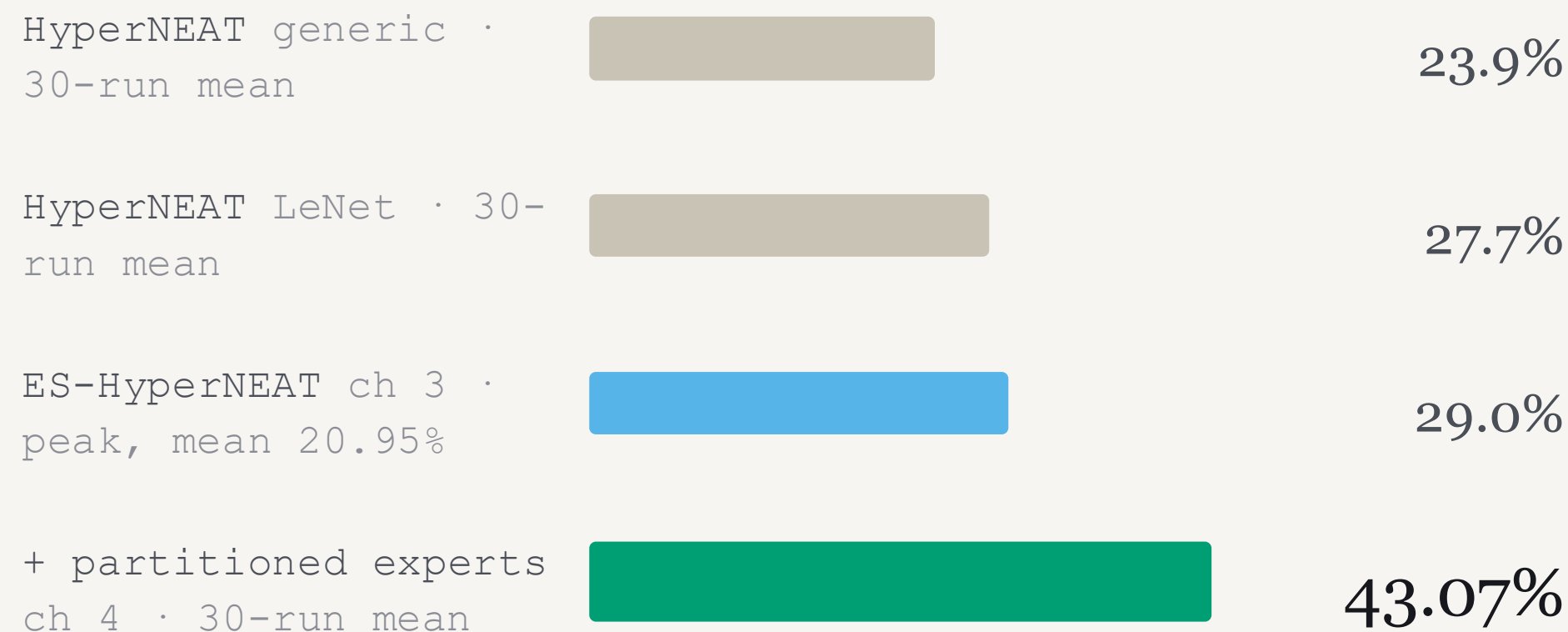
## IN RL

Irrelevant. a different task entirely

## AND THE COST SIDE

Reached with **~100× fewer neurons**, and the substrate was *discovered*, not designed.

## MNIST, WITHIN NEUROEVOLUTION

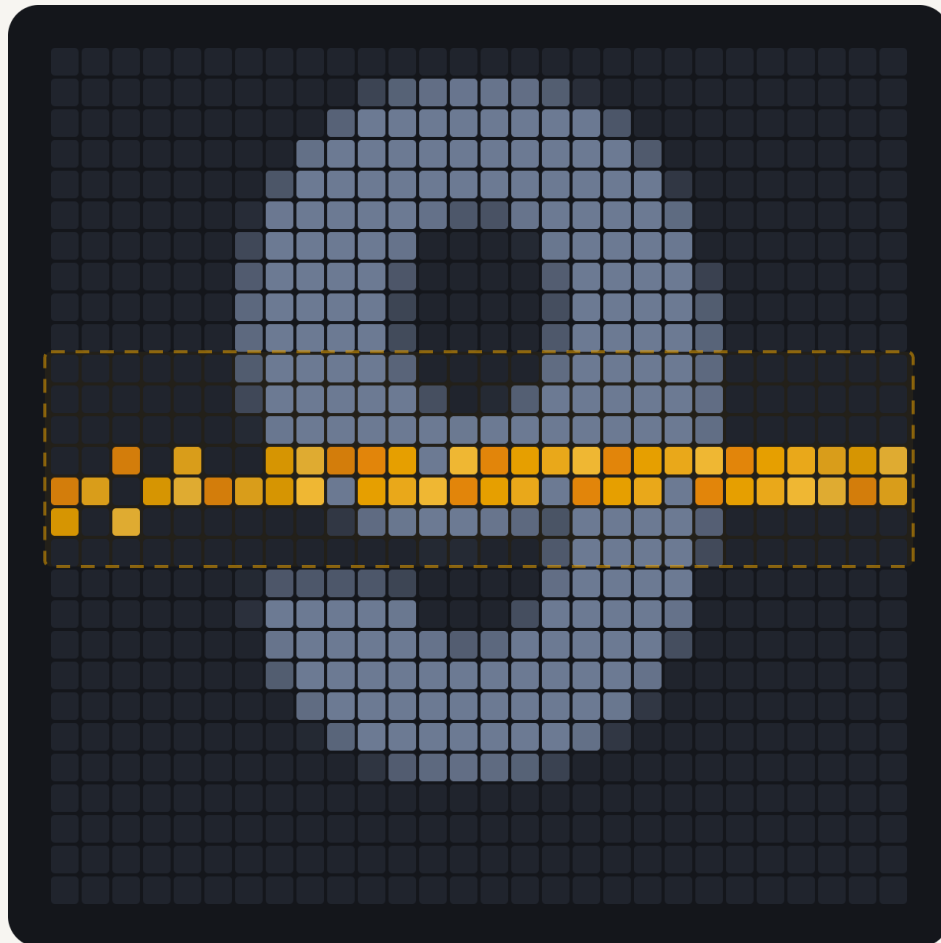


Mean-to-mean, the ch 4 row is the only one that clears both baselines. Both ES-HyperNEAT rows discovered their own substrate; the two baselines were hand-designed. Peak run: 50%.

The contribution is the **diagnostic insight** (the bias blocked 96% of the input) and the **architectural principle** (decomposition recovers it). Not competitive accuracy, and I won't claim otherwise.

# Three mechanisms compound into collapse.

ONE MEASURED RUN · ACTIVE INPUT  
PIXELS



TWO COUNTS, ONE SEARCH

48 px · 6.1% · this figure  
one of the search's best networks

28 px · 3.6%  
coverage convention, used elsewhere

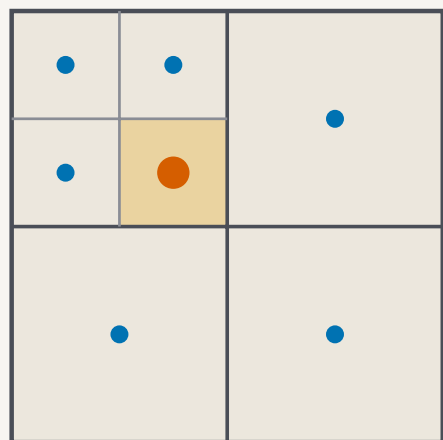
- 1 Coordinate symmetry  
The quadtree starts at the origin because the optimal start is unknown *a priori*.
- 2 Symmetric CPPN primitives  
Gaussian, sigmoid, tanh all peak their output variance near the center.
- 3 Variance-driven subdivision  
Resolution is added exactly where variance is high → all of it, on the midline.

**Not a bug in one run.** All 30 replications land inside rows 10–16, and it holds across every viable hyperparameter, not a configuration I happened to pick. Structural, not a tuning artifact.

A 10-line receptive-field plot would have caught this on day one.

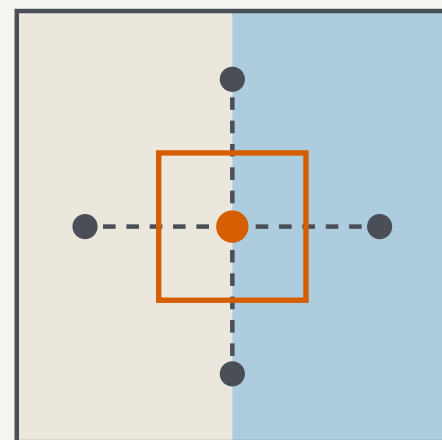
# Where should the hidden neurons go?

## 1 · DIVIDE



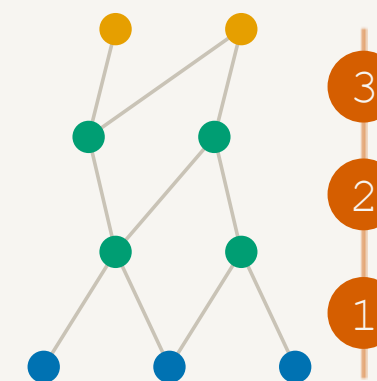
From one **anchor node**, query the CPPN at 4 quadrant centers. If their **variance** exceeds  $\theta_{\text{div}}$ , that region holds structure, so split and recurse.

## 2 · KEEP THE BANDS



Probe 4 neighbors a cell-width out. Keep the leaf only where weight changes on **both sides** of an axis:  $\text{band} > \theta_{\text{band}}$

## 3 · THREE PHASES



Repeat inputs→hidden, hidden→hidden ( $\times L$ ), hidden→outputs. Then prune every node off an input–output path.

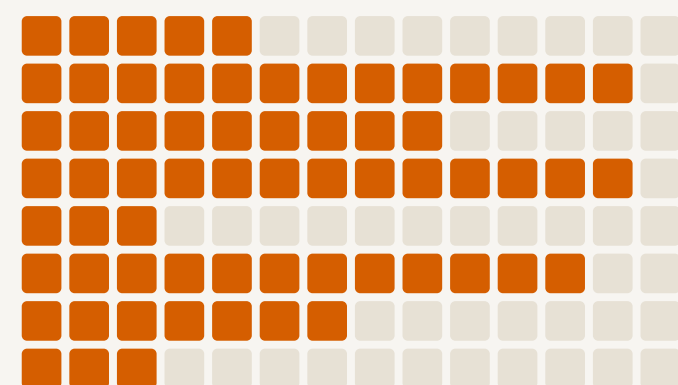
**The catch:** which cells get split depends on values you only learn by querying. Every genome grows a **differently-shaped tree**, so there is no fixed-shape array anywhere in it. That, not the arithmetic, is what cannot be batched.

# Structural, not an implementation complaint.

## WHY SIMT CAN'T RUN IT

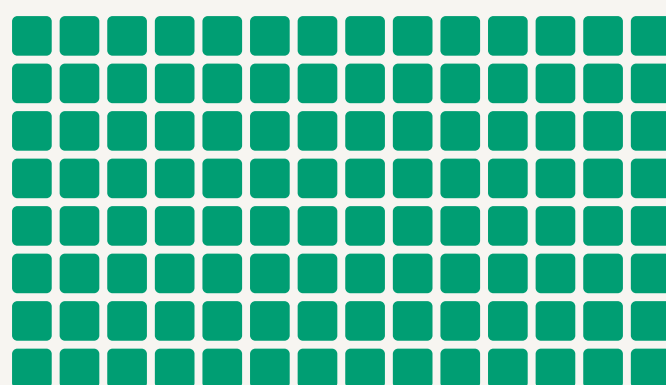
A warp of 32 threads executes **one instruction at a time** .  
Every genome builds a **different tree** , so lanes finish at different depths, and the whole warp waits for the slowest.

### QUADTREE



schematic: lanes idle at different depths; the warp waits for the slowest

### EMR



schematic: every lane busy at every step, by construction

HSHG spatial hashing was tried as an alternative. It over-discovers (29–63 nodes where 1–2 suffice). The real problem is **topology uniformity**, not speed.

## THE EVIDENCE

$F(6,449)=87^{\text{depth}}$  as the runtime factor

$p < 10^{-71}$        $\eta^2 = 0.54 \cdot R^2 = 0.95$

21,845      candidate positions at depth 7

1.65×      cumulative ceiling of 4 strategies (reference config)

### AGAINST WHAT WAS AVAILABLE

1.65× vs 500×

Full population vectorization over 500 individuals.

**Three orders of magnitude** left on the table, no implementation reaches it.

Scope: SIMT hardware. Dataflow / FPGA untested.

# Evaluate a fixed grid, then mask by variance.

## STEP 1 · GRID

Fixed multi-resolution lattice  
Every candidate position, all depths,  
defined up front and identical across  
the population.

## STEP 2 · EVALUATE

One batched CPPN pass  
vmap + JIT → a single matrix op over  
the whole population, on any  
backend.

## STEP 3 · FILTER

Mask where  $\text{var} > \theta$   
Same criterion as the quadtree,  
applied post-hoc. Sparsity kept;  
shape uniform.

## THE SHAPES · ONE GENERATION

|          |                         |                         |
|----------|-------------------------|-------------------------|
| grid     | [ positions × coords ]  | fixed before any run    |
| CPPN out | [ pop × pos × outputs ] | one batched pass        |
| variance | [ pop × pos ]           | one number per position |
| mask     | [ pop × pos ]           | all that differs        |

## WHAT THE FIXED SHAPE BUYS

**Depth 13** by streaming across devices: **memory**  
scaling, not speed. Multi-GPU costs 1.5–1.7×, still  
~20× > CPU.

**Extensibility:** per-node activation, aggregation and  
modulation are extra **output columns** , no traversal  
to change.

**The cost:** we evaluate positions the quadtree never  
visits. At shallow depths the gap is *marginal*; the win  
arrives with depth.

**Every shape is known before evaluation.** Genomes differ in mask  
*contents* , never in array dimensions, what the quadtree could not  
offer.

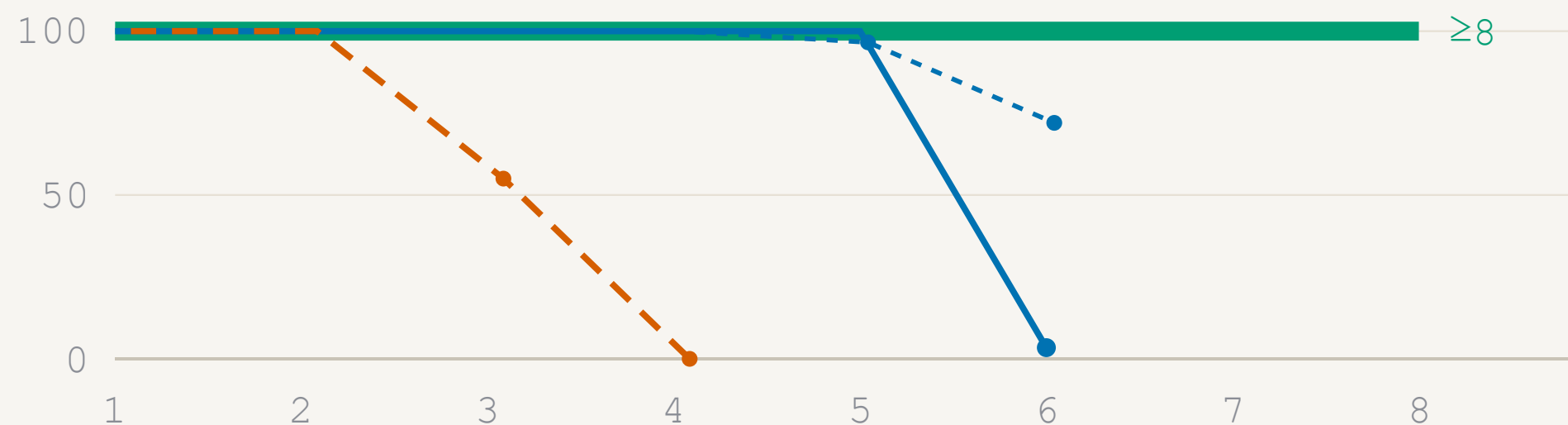
# New biology is a new column.

Every neuron is a row in a tensor. New biology is a **new CPPN output channel** read by the forward pass, not a new algorithm.

|   |                                                                                              |                                                                                                   |
|---|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| W | <b>connection weight</b> the only channel ES-HyperNEAT had                                   | the baseline substrate                                                                            |
| f | <b>per-node activation</b> selected from an 18-function palette                              | Oscillatory functions solve parity at any bit-width; every monotonic one fails.                   |
| Σ | <b>per-node aggregation</b> sum, product, max, evolved alongside <i>f</i>                    | The palette itself can be <b>meta-evolved</b> across generations.                                 |
| r | <b>receptor densities</b> a 4-vector per neuron, gated by a per-task neurotransmitter signal | One substrate, many behaviors, and a <b>multi-task barrier</b> that neuron-type diversity breaks. |

# Where the edges are, precisely.

SUCCESS RATE vs PARTS COMPOSED



■ continuous · additive mean — continuous · multiplicative

— boolean · composition — boolean · + gating

**One operator scales, one dies at 4.** Same grids, same mappers, same budget, so the limit sits in the **composition operator**, not the substrate.

**0.53–0.80** zero-shot fitness  
(vs the 0.95 bar)

Individual mappers learn task-specific routing.

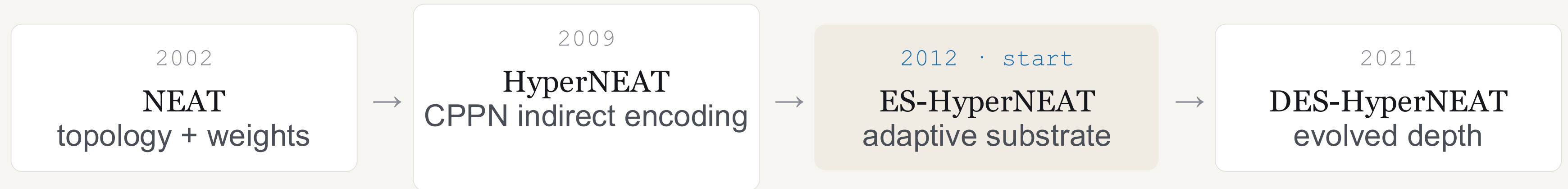
**0.3%** latent solvers (vs  
<0.01% random)

The signal is population-level. Warm-starts converge 45–79× faster at 3 grids, 223–351× at 5.

STILL UNDEMONSTRATED

mapper transfer · spontaneous specialization ·  
super-additive composition

# The lineage, and the closest neighbors.



## Closest: CoDeepNEAT

Co-evolves modules + blueprints, but gradient-trained and fixed-topology. EMR alone gives spatial organization *and* per-node feature evolution at practical cost.

## Continual learning

EWC, Progressive Nets, PackNet assume gradients + fixed topology, so none apply to evolved substrates. Frozen-grid reuse is the native answer.

~14,500 runs · ≥2,100 compute-hours.

| CHAPTERS               | RUNS   | HOURS  | HARDWARE                         |
|------------------------|--------|--------|----------------------------------|
| Optimization + experts | ~5,900 | ~900   | Distributed CPU (13–14 machines) |
| Quadtree proof         | 456    | ~50    | M4 Max (CPU) + RTX 2080 Ti (GPU) |
| EMR-HyperNEAT          | ~5,000 | ~150   | M4 Max (CPU) + RTX 2080 Ti (GPU) |
| GEENNS prototype       | ~3,100 | ≥1,015 | M4 Max (CPU)                     |

# From here.

## Distill the signal

Population-level composition → a single mapper, via attention-based gating (MoE-style).

## Hybrid NAS

EMR discovers topology; gradient descent trains the weights inside it.

## Multi-modal & industrial

Grids per modality. Job-shop, portfolio, control: real high-dimensional search.

## DEMR, deep substrates

Multi-layer eager discovery; directly targets the Two-Spirals expressiveness failure.

## Harder benchmarks

CIFAR-10 (color, hierarchy) and ARC (explicit compositional generalization).

## The artificial-life program

Measure emergence with training-data-free metrics, pre-registered against nulls. B14 · B15

# Four papers + a preprint in the thesis; three built on it.

## IN THE THESIS

|    |                                                      |           |
|----|------------------------------------------------------|-----------|
| P1 | TPE hyperparameter optimization, GECCO '24 Companion | published |
| P2 | Early-stopping thresholds, IEEE WCCI 2026            | published |
| P3 | Breaking the central bias, ICPR Biomap 2026          | published |
| P4 | The quadtree bottleneck, arXiv                       | preprint  |
| P5 | EMR-HyperNEAT, GECCO Companion '26                   | published |

## BUILT ON THIS FOUNDATION

Per-node activation functions · published

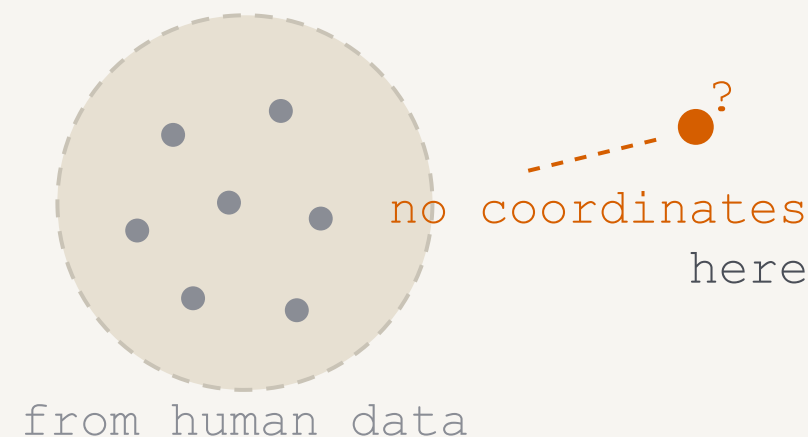
Bio-inspired palette meta-learning · published

Neuromodulation for multi-tasking · published

First author on all five thesis papers: conceived the questions, designed the algorithms & protocols, built the codebase, ran the analyses, wrote the manuscripts. Arthur Gygax co-authored P2 & P3.

# How would you prove a behavior is new?

## THE CIRCULARITY



Scoring novelty as distance in a foundation model's embedding space measures a space built from *us*. Behavior outside it cannot be told from noise.

## INSTRUMENTS THAT NEED NO TRAINING DATA

Bedau evolutionary activity

Sorts long-run dynamics into absent, bounded, unbounded, against a neutral shadow model.

MODES hallmarks

Change, novelty, complexity, ecology, under a persistence filter.

Effective information

A causal-emergence probe, exploratory and feasibility-gated.

All declared against neutral and random nulls, with a degeneracy filter so drift alone cannot pass for adaptation.

**The falsifiable bet:** foundation-model novelty will **not** track training-data-free novelty. A strong correlation kills the premise, which is exactly why it is worth registering first.

# Measuring behavior that was never learned from us.

Proposed 24-month program · UCD Natural Computing, with Michael O'Neill

## O1 · INSTRUMENT

Build the toolkit and benchmark suite. Every metric, null and threshold frozen publicly *before* any run that could tune them.

## O3 · DO MAPPERS GENERALIZE?

Distill the population-level signal into one mapper, encapsulating recurring routing motifs as an evolved grammar.

## O2 · DOES SPECIALIZATION EMERGE?

My grids were hand-assigned. Test whether roles arise on their own: mutual information with a geometry-fixed grid index, above a label-shuffled null.

## O4 · OPEN ASSET

Toolkit, pre-registrations, replication data and negative results, released. One consumer GPU at <30% duty.

**The through-line:** this thesis names two signatures it did **not** demonstrate, emergent specialization and robust zero-shot generalization. They are O2 and O3. A clean negative is a primary, publishable outcome.

# Because the vision was arithmetically impossible first.

THE TRAJECTORY · APPENDIX A

|               |                                                                                        |
|---------------|----------------------------------------------------------------------------------------|
| 2020          | 4 architectures for grounded reasoning → all assume a fixed graph                      |
| 2022          | fuzzy logic dies combinatorially; cortical columns point to self-organizing substrates |
| 2023 · Dublin | 2,013 trials. Two side observations outlive the headline                               |
| 2024          | 48 pixels → partitioning. 2,013 trials/CPPN → the pruner                               |

THE FORCING NUMBER

GEENNS at a *modest* 5 grids, 10 tasks needs 34 CPPNs. At 2,013 trials each: 68,000 trials, roughly **7 years** of continuous CPU.

The vision was not conceptually hard. It was unaffordable. So the quadtree chapters are its **precondition**, not a detour from it.

**And the order was forced, not planned** , which is why Appendix A documents the pivots instead of presenting a tidy retrospective plan.

# EMR removed barrier 3 and left barrier 2 standing. The experiment was run for me.

## THE DECISIVE TEST

“A fast algorithm that produces a centrally biased substrate is still a centrally biased substrate; it merely arrives at its inadequate solution faster.” — Ch 1

EMR applies the **same** variance criterion. Speed was never the representational problem. **Partitioning** fixed barrier 2, one layer above the algorithm.

## THE HONEST NUANCE

|                   |                                                                                     |
|-------------------|-------------------------------------------------------------------------------------|
| as a narrative    | not independent: they form a progression: ceiling, then structural, then unscalable |
| as repairs        | independent, search problem · architectural problem · algorithmic problem           |
| the thesis' words | “neither solution alone is sufficient; both are necessary”                          |

If it were one barrier in three masks, one kind of fix would have cleared all three. None did.

■ Q3 · CH 1 “WHY EXACTLY THREE? AND CH 7 NAMES A DIFFERENT THREE.”

# Two targets, two decompositions. The shared root is why it got two chapters.

## THE TWO TRIADS

|         |                                                                                     |
|---------|-------------------------------------------------------------------------------------|
| Ch 1    | <b>practicality:</b> configuration · representation · computation                   |
| Ch 7    | <b>composition:</b> computational · expressiveness · multi-task                     |
| overlap | computational appears in both: the shared root                                      |
| timing  | expressiveness and multi-task only become visible <i>once</i> the substrate is fast |

## AND WHY 4 QUESTIONS FOR 3 CLUSTERS

Ch 4 answers two: it **diagnoses** the representational collapse (TRQ2) and supplies the **decomposition** remedy (TRQ4). Diagnosis and cure in one chapter, counted separately.

On “complete”: complete relative to a stated goal, making adaptive substrate neuroevolution practical, and not in any absolute sense. I defend it only in that scoped reading.

**So the lists differ because the questions differ**, barriers to a *practical algorithm*, then barriers to *compositional intelligence*.

■ Q4 · CH 1 “BARRIERS 1 AND 2 WERE SOLVED ON CPU, NEVER COMBINED WITH EMR.”

True, and stated in the limitations. The dependency runs through the pruner.

CONCEDED

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| Ch 3–4             | CPU-only. EMR did not exist when they ran.                  |
| partitioning + EMR | named in Ch 8 as remaining <i>engineering</i> , not science |
| so no              | there is no single unified system to demo                   |

WHAT IS ACTUALLY INTEGRATED

The early-stopping result cut campaign cost enough that the **thousands of runs in the later chapters became possible at all** . Chapter 3 paid for Chapter 6.

The integration is temporal and economic rather than architectural, which is what a **diagnostic program** looks like, as opposed to a system release.

**The claim is intact:** three structural barriers found, characterized, and broken. Combining them is the next commit, and I would rather say that plainly.

# The age is evidence. The missing baseline is a stated limitation.

## WHY THIS LINEAGE

ES-HyperNEAT is still the **only** line that removes the human from substrate design. HyperNEAT makes you place the neurons; everything since either fixes the substrate or trains it by gradient.

It stalled for 14 years. My thesis argues it stalled *on the bottleneck I removed* , so the age supports the diagnosis. DES-HyperNEAT (2021) extends the same foundation.

## THE MISSING CONTROL, OWNED

No gradient-trained control exists in these experiments. The thesis lists it as a limitation affecting **every chapter**.

Why: gradients need the topology to already exist, and topology discovery is what is being measured. Discovered substrate against hand-designed-plus-backprop confounds encoding with optimizer.

**So 99% vs 43% is context, never a result** , and the hybrid (EMR for topology, gradients for weights) is the honest way to settle it. Future work.

# NAS in the broad sense. Not in the narrow one.

THE MODERN FORMULATION · ELSKEN ET AL. 2019

|                        |                                       |
|------------------------|---------------------------------------|
| search space           | the set of architectures considered   |
| search strategy        | how to explore it                     |
| performance estimation | <b>gradient-trains</b> each candidate |

What it returns is a **blueprint awaiting training** , retrained from scratch. So modern NAS implies *training after search*, by construction.

THE BOUNDARY CASE I CONCEDE

Evolutionary methods *do* appear inside modern NAS as one search strategy, but there the weights are still gradient-trained. **NAS names the problem; a search strategy answers only where to look next.**

WHERE THE NEAT LINEAGE BREAKS IT

|                    |                                                |
|--------------------|------------------------------------------------|
| topology + weights | carried together in <b>one genome</b>          |
| gradient training  | none, at <b>any</b> stage                      |
| what it returns    | a <b>functioning network</b> , not a blueprint |

It does not slot into that decomposition: it **breaks the decomposition itself** . Already encoded in my comparison table’s weight-optimization row.

AND NOT “A NAS THAT DOES NAS”

The CPPN doesn’t *search* for a substrate: it **develops** one. Architecture search over architecture **builders** : one level above the formulation, which is what makes the space *open* where cell-based spaces are fixed.

■ Q7 · CH 3 “IS THE 41.6% SAVING MEASURED, OR PROJECTED?”

Measured. The combined pipeline is the projected part, and labeled so.

MEASURED

|                 |                                       |
|-----------------|---------------------------------------|
| 41.6%           | generation cost reduction             |
| 50.6%           | trial pruning ratio                   |
| $F_1 = 0.872$   | pruner validation                     |
| TOST $p < .001$ | equivalent quality, not a null result |
| 64% fewer gens  | vs Hyperband, $d = 1.884$             |

PROJECTED, SAID SO IN THE CHAPTER

TPE + pruner end-to-end was **never validated as one system** . Combined figures are derived analytically from the measured savings, under an explicit no-interference assumption.

Calibration: both pruner parameters ( $G^*=3$ ,  $T^*=0.140$ ) come from **MNIST alone** . Hyperparameters transfer MNIST→Fashion-MNIST, not MNIST→XOR.

A projected figure presented as measured should sink a thesis, so the chapter separates them, and names end-to-end validation as the next step.

# The size is not the problem. The shape is, and the interpretability is worse.

## THE SHAPE, NOT THE SIZE

|                   |                                                             |
|-------------------|-------------------------------------------------------------|
| 3B configurations | merely large, 16-parameter subspace                         |
| median = minimum, | so $\geq 50\%$ of trials sit exactly on the guessing floor  |
| 16.6% never move  | off the 10% floor, across all 2,013 trials                  |
| the consequence   | hand-tuning does not give a worse result. It gives nothing. |

## THE REAL REASON IT COMES FIRST

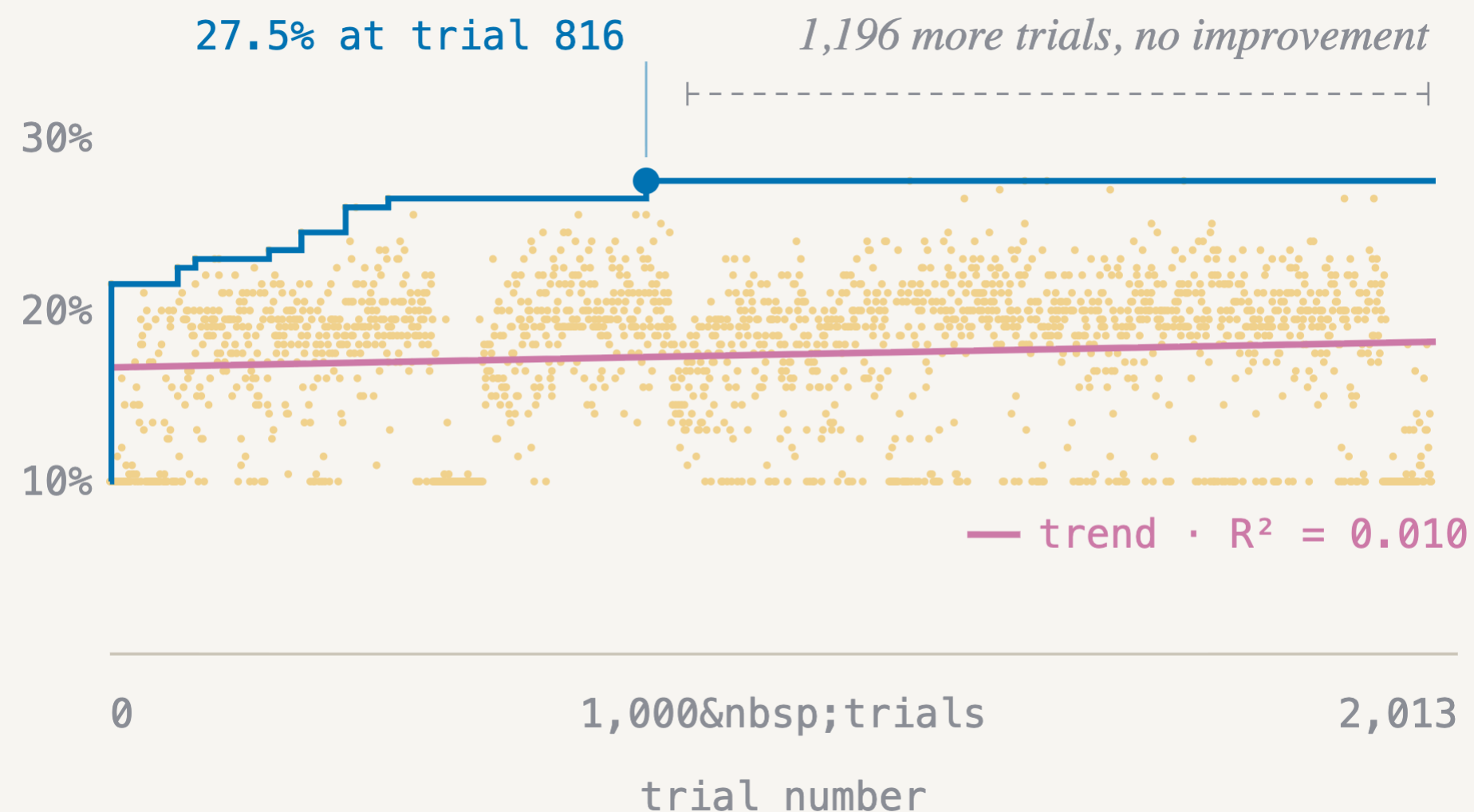
Prior ES-HyperNEAT work used **hand-selected** hyperparameters with no systematic optimization. So no published number could be attributed to the algorithm rather than to a lucky draw.

Until someone paid the search cost, the literature was **uninterpretable, including its ceiling** . That is why the least fundamental barrier had to be first.

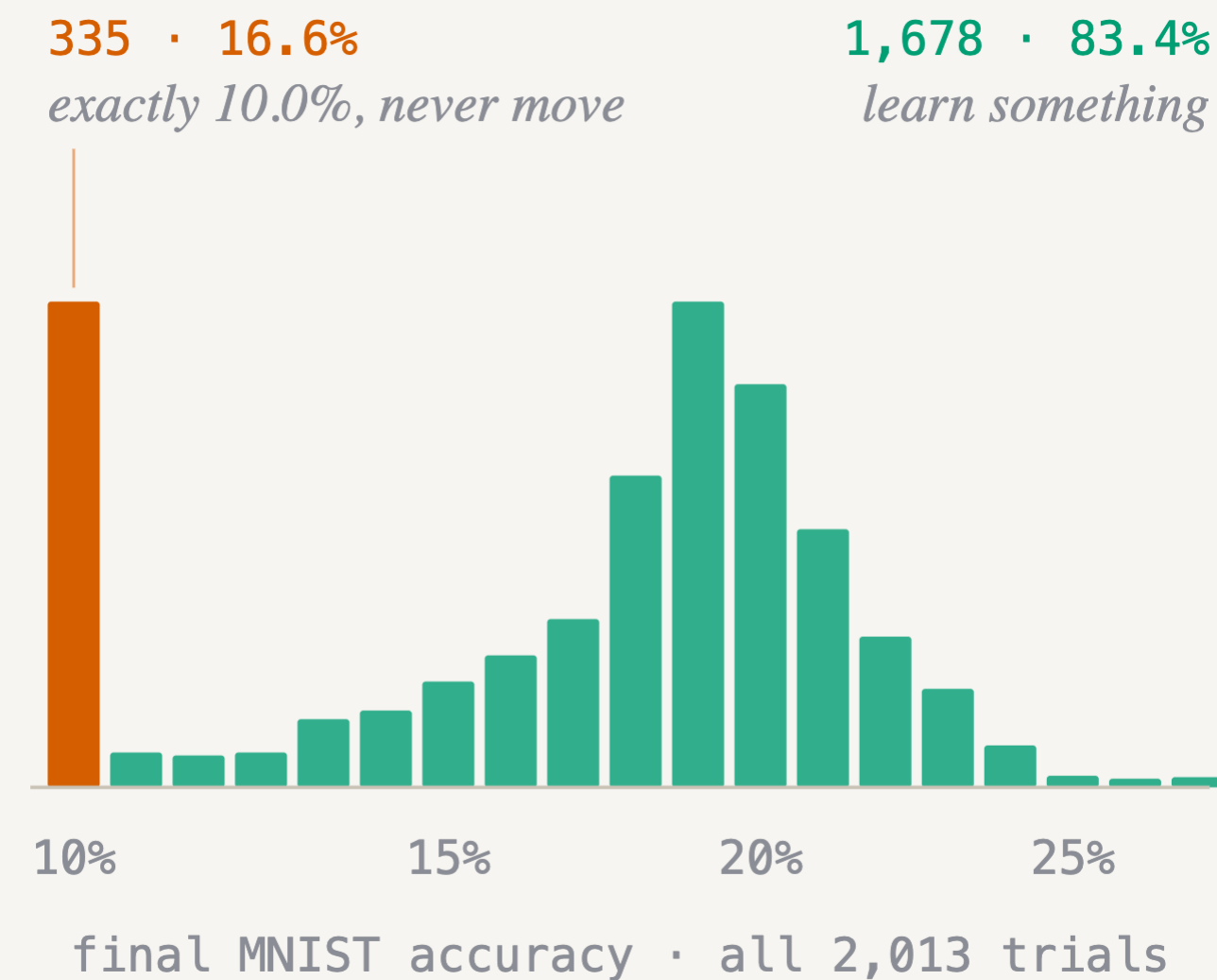
**Yes, it is practical rather than fundamental** , Ch 1 says so. It is also the precondition for trusting anything downstream.

# Eight improvements in 2,013 trials. Then nothing.

SEARCH PROGRESS · MNIST · TPE



THE DISTRIBUTION THEY FORM



**60% of the budget bought nothing**, which is what makes the ceiling structural, and what the generation-3 pruner exploits.

# Pruning framed as binary classification.

Schedule-based pruners drop a fixed fraction at preset budgets, blind to how *this* algorithm fails. Instead: read its own dynamics. **One feature, one generation, one threshold.**

## SUCCESS-DEFINITION SWEEP

| DEF. | RULE        | RECALL | SAVINGS |
|------|-------------|--------|---------|
| 50th | (13, 0.145) | 0.52   | 11.4%   |
| 75th | (4, 0.143)  | 0.86   | 38.2%   |
| 80th | (3, 0.140)  | 0.90   | 41.6%   |
| 90th | (3, 0.140)  | 0.94   | 41.6%   |

The 80th maximizes **recall** at the largest savings. The 75th has a higher  $F_1$  but prunes later and keeps fewer winners.

## THREE CHOICES, ALL MEASURED

|                  |                                                                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| which metric     | <b>median</b> best fitness. Last-best fails: its 80th percentile sits <i>on</i> the 0.1 floor, nothing to threshold. Median vs mean: no difference (p>.05, d<0.2) |
| which generation | <b>3</b> . Beats gen 2 (McNemar p = .022), ties gen 4 (p = .375) → take the earlier. $F_1$ .819 / <b>.872</b> / .869                                              |
| which threshold  | <b>0.140</b> , where $F_1$ peaks. Winners and losers separate by gen 3 with $\Delta \approx 0.037$                                                                |

**The rule:** kill any trial whose median best fitness at generation 3 is  $\leq 0.140$ . The methodology transfers to any algorithm whose failures are diagnosable early. The numbers do not: they are re-derived per task.

# Every pruned trial was resumed to reveal its true outcome.

Kill a trial and you never learn what it would have done. So on **180 held-out trials** , the pruned ones were run to completion anyway, and the confusion matrix is scored against ground truth, not against a guess.

|                      |                      |                       |                      |
|----------------------|----------------------|-----------------------|----------------------|
| 75<br>true positives | 8<br>false negatives | 14<br>false positives | 83<br>true negatives |
|----------------------|----------------------|-----------------------|----------------------|

| THE CLASSIFIER             |                                      |
|----------------------------|--------------------------------------|
| recall                     | <b>0.904</b> : keeps >90% of winners |
| F <sub>1</sub> · precision | 0.872 · 0.843                        |
| MCC · accuracy             | 0.757 · 0.878                        |

| THE PAYOFF     |                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------|
| cost, same 180 | 3,060 → <b>1,786</b> generations (−41.6%)                                                                                     |
| quality        | TOST vs unpruned, margin ±0.02, diff −0.006, <b>p &lt; .001</b> : equivalence <i>claimed</i> , not just absence of difference |
| vs Hyperband   | 10,656 vs 29,574 generations: <b>−64%</b> , higher mean top-5% fitness (d = 1.88)                                             |

# No. It is calibrated for random exploration.

Replayed counterfactually on the **824 elite trials** of the converged GECCO’24 campaign, the rule still separates good from bad, but it becomes too strict.

| WHAT STILL HOLDS |                                    |
|------------------|------------------------------------|
| separation       | p < .001: the signal is real       |
| kept vs pruned   | mean 0.187 against 0.150           |
| savings          | 73.2% fewer full-trial equivalents |

| WHAT BREAKS    |                                                                            |
|----------------|----------------------------------------------------------------------------|
| recall         | <b>90.4% → 31.1%</b> on top-20% trials                                     |
| the best trial | 0.275, a <i>slow starter</i> , is killed. Kept peak caps at 0.245          |
| why            | slow starters are rare under random init and common in an elite population |

**Which is the next step:** co-evolve the threshold alongside the search so it relaxes as the population improves: the BOHB and DEHB direction.

# Scoped, structural, and testable.

## ESTABLISHED

Three compounding mechanisms, all in the **encoding's geometry** : normalized coordinates, origin-symmetric primitives, variance-driven subdivision. Reproduces across all 30 replications and every viable configuration.

The scoping cuts both ways, stated in the thesis: direct encodings, or generators that are *not* origin-symmetric, should not show this pathology.

## THE OPEN TEST

The falsification experiment is specified, not hand-waved: swap in **asymmetric primitives** , shifted sigmoids, learned radial centers, and ask whether the basin survives when nothing peaks at the origin.

Untested; future work. Partitioning fixed the *symptom* one level up, at the architecture: a fix that works whichever mechanism dominates.

**So the claim is scoped:** structural within ES-HyperNEAT's geometry, with the primitive-swap experiment named as the root-cause test.

# 13 is where receptive field meets search capacity.

WHY 13, SWEPT 1 TO 17, N=30

|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| 784 / 13 ≈ 60 px            | the largest domain one population explores effectively           |
| finer (N>13)                | too few pixels to tell a 1 from a 7                              |
| coarser (N<13)              | the central bias re-emerges <i>inside</i> each segment           |
| 43.07% at N=13<br>· see Q15 | coverage peaks elsewhere (85% at N=8, 14), quality, not quantity |

## THE ENSEMBLE OBJECTION

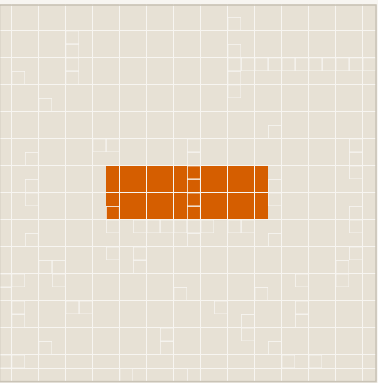
Krogh & Vedelsby: ensemble error = mean individual error – a **diversity term** that vanishes when members converge on the same features. 13 monoliths all collapse to the same band, so they share blind spots.

Individual experts stay near baseline at every N, so the gain is diversity rather than better experts. And N=1 reproduces the original baseline at  $p = .502$ .

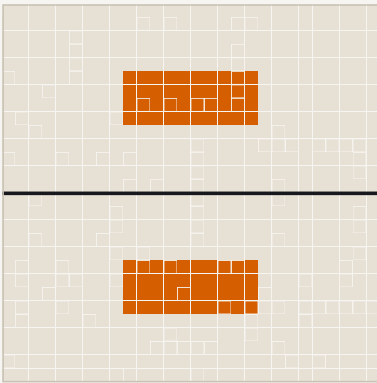
**The honest gap:** my gating is static. A learned gate might find better assignments: I traded that for a search space evolution could afford.

# 48 px was never a capacity limit. It was a satisficing point.

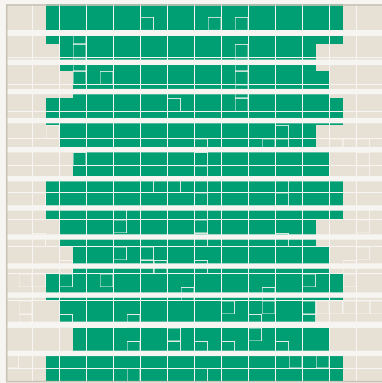
Partitioning is **1-D linear slicing** : expert k takes pixels  $[k \cdot 784/N, (k+1) \cdot 784/N)$ , which in a 28×28 image is whole rows. A thin band, not a square.



1 expert  
48 of 784 px



2 experts  
two central  
clusters



13 experts  
~73% of the  
input

## WHY THE MONOLITH STOPS AT 48

no pressure      the central cluster already suffices, so the search **settles**

a band can't      rows 0–9 have *no center to retreat to*

geometry      the bias is radial; a 28×9 band has no periphery

## WIRED ≠ EXPLOITED

N = 3      ~250 wired, ~55 exploitable: **no leverage**

N = 13      ~60 wired, ~60 exploitable: all of it earns its place

N = 8, 14      85% coverage, lower accuracy: it is *free* to grow

**A wording I would correct:** Ch 4 calls 48 px the maximum receptive field the algorithm can manage, but my N=3 data contradicts that. The ceiling is on **effective dimensions in a fixed budget**, not connection count.

# It does, and the thesis says so in the same sentence.

## WHAT IS SCOPED

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| the reference      | XOR at depth 2, pop 500, task-, depth- and population-dependent |
| ablation, pop 1000 | O2+O3 alone reach 1.73×; O4 alone only 1.01×                    |
| hardware           | SIMT only. Dataflow and FPGA untested.                          |

## WHY THE NEGATIVE HOLDS

At deeper substrates JIT overhead **inverts the gain** : the vectorized quadtree becomes *slower* than baseline. The ceiling turns negative exactly where you need it.

And the untunable comparison: full population vectorization over 500 individuals would in principle give 500× . 1.65 against 500 is three orders of magnitude.

**Which is the actual conclusion:** the barrier is population-level heterogeneity, not efficiency within an individual.

# Same queries. The dependency chain is what left.

## THE ACCOUNTING

Sequential:  $O(4^D)$  chained queries,  $O(H^2)$  connectivity. EMR: the same counts over  $P$  lanes,  $O(4^D/P)$  elapsed.

|                  |                                                   |
|------------------|---------------------------------------------------|
| depths 1-3       | the gap is <i>marginal</i> : the thesis' own word |
| depth 4          | crossover, 1.5×                                   |
| depths 5 · 6 · 7 | 12× · 23× · 34×                                   |
| memory           | the real price of eager; streaming → D13          |

## WHY NAIVE EAGER FAILS

**HSHG** pre-allocated, batched, thresholded, and over-discovered 29-63 nodes where the quadtree finds 1-2: overlapping neighborhoods let one feature fire many positions.

EMR re-couples variance to the hierarchy: 4 **non-overlapping** children, parent-conditioned mask. The quadtree's selectivity, inside fixed-shape tensors.

JAX port of the quadtree: selectivity kept, 1.65× ceiling kept. **EMR is the design that gets both.**

**The point:** the quadtree's cost was never how many queries. It was that each one waited for the last.

# Exact under matched settings. Divergent by choice in the defaults.

## THEOREM · SET EQUALITY UNDER MATCHED SETTINGS

Under identical  $\theta_{\text{div}}$ ,  $\theta_{\text{band}}$ , and substrate-selection rule, EMR and ES-HyperNEAT discover **identical position sets** and produce **identical connection sets**. Proof: both admit a position iff every ancestor’s 4-child variance clears  $\theta_{\text{div}}$ .

## THE HONEST CAVEATS, SHIPPED DEFAULTS DIFFER

|                     |                                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| substrate selection | ES-HN keeps leaves; EMR keeps the multi-res union → strict superset in practice |
|---------------------|---------------------------------------------------------------------------------|

|            |                                                               |
|------------|---------------------------------------------------------------|
| thresholds | 0.03 raw vs 0.5 scaled → EMR more permissive on the same CPPN |
|------------|---------------------------------------------------------------|

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| root level | always active in EMR → valid substrates where the quadtree cold-starts to none |
|------------|--------------------------------------------------------------------------------|

**So:** “identical” is a theorem, not a slogan, and every deliberate divergence from its premises is documented and measured.

# 15 problems, 4 domains, including where it fails.

HOLDS · RL CONTROL & VISION

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| CartPole      | 100% solve, N=30                                                |
| Acrobot       | 100% solve, N=30                                                |
| Retina (L/R)  | 100% solve, N=60                                                |
| MountainCar   | 43%, sparse reward: an exploration problem, not a substrate one |
| CCM financial | speedup transfers: 5.5×                                         |

FAILS · REPORTED AS RESULTS

**Regression is a domain limitation.** CCM  $R^2$  0.200 vs NEAT  $\approx 0.41$ , conventional ML 0.42–0.44 . CalHousing 0.319 · Earnings-Price 0.451.

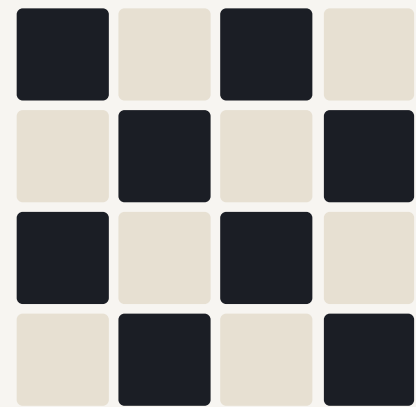
Why: weight-sharing yields smooth connectivity built for *boundaries* , not surfaces; substrates carry only 20–50 hidden nodes; no per-node function diversity applied yet.

**Two-Spirals: 0%.** An expressiveness limit, named in the thesis.

**The claim:** feasibility across domains and a speedup that transfers, not competitiveness. The failures are primary results, not footnotes.

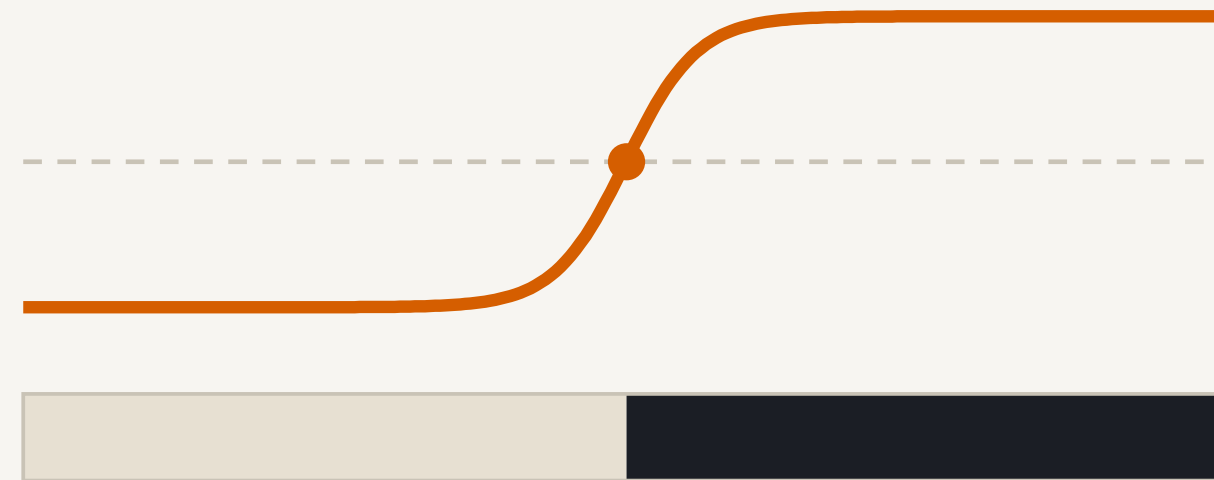
# Parity needs a boundary that *repeats*.

Flip any single bit and the answer flips. So the decision surface is a checkerboard, and a neuron can only draw it if its activation function **crosses zero more than once**.



what parity asks  
for

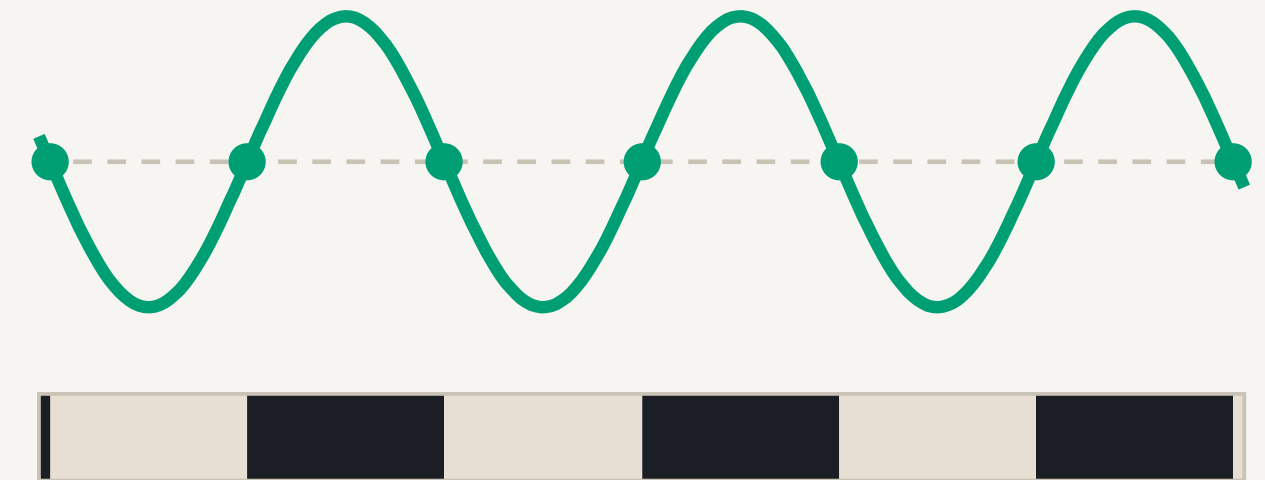
MONOTONIC · tanh, sigmoid



Crosses zero **once**. The space splits in two, and no amount of training adds a third region.

**0%** every function, every bit-width

OSCILLATORY · sin



Crosses zero **again and again**. The space stripes for free, which is exactly the checkerboard.

**100%** at any bit-width tested

Never vs always, decided by one function, and EMR lets each neuron discover it.

# The null was paid for. The solvers are the same individuals.

## WHY IT’S SIGNAL

|                 |                                                                              |
|-----------------|------------------------------------------------------------------------------|
| the null        | random init at pop 3000, 10× budget: <0.01% vs ~0.3%                         |
| Jaccard overlap | solver sets across targets: median 1.0, mean 0.88: one shared representation |
| speedups        | 45–79× (3-grid) · 223–351× (5-grid)                                          |

## WHERE IT WEAKENS, MAPPED

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| input permutation | transfers cleanly                                              |
| gate substitution | NOR→XNOR: 2.06× only; 4/29 zero-shot at f≥0.99                 |
| additive operator | 12/14 (5-gate) · 59% (6) · 86% (7), tracks structural distance |

So the encoded signal is partly compositional, partly distributional. Presence + recoverability + a mapped boundary.

**Open, by design:** extraction of that signal into a single mapper, O3 of the next program (B15).

# The non-significant part is labeled. The significant part carries the claim.

## WHAT’S SIGNIFICANT, AND WHAT ISN’T

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| 4-grid success gap     | 100% vs 87%: NOT significant, Fisher $p=0.112$ , stated in the thesis          |
| 4-grid speed inversion | 8× slower: Mann-Whitney $p=3.2\times 10^{-8}$ , $r=0.87$                       |
| the 4 failures         | plateau at $f=0.949\text{--}0.950 \rightarrow$ capacity limit, not slow search |
| 5-grid collapse        | Fisher $p<10^{-17}$ vs composition                                             |

## THRESHOLDS vs THE FLOOR

Class imbalance: an all-zeros predictor already scores  $0.982$  at 5-grid, so the bar is  $f\geq 0.99$ . At 6-grid: floor  $0.9912$ , bar  $0.995$ .

And the anomaly, flagged by me: mono-tanh recovers to 27% at 4-grid because NAND is linearly separable, 8 hard-won solves, median 437 gen,  $p=0.002$  vs chance. Task structure interacts with capability at every scale.

**Bottom line:** the cliff survives the scrutiny built to kill it: the claim rests on the inversion and the 5-grid collapse, not on any single cell.

# Freezing is the forgetting fix. The growth is quadratic, and I state it.

## WHY FROZEN

Evolve a substrate for task A, keep evolving for B, and A’s weights are overwritten. That is catastrophic forgetting in evolutionary terms.

Freezing makes it **impossible by construction** : the specialist stops changing, everything task-specific lives in the mappers. Accumulation instead of replacement.

## THE GROWTH, OWNED

|                  |                                                                 |
|------------------|-----------------------------------------------------------------|
| $4+k(k-1)/2+2T$  | 13 CPPNs (3 grids, 3 tasks) · 34 (5, 10) · 89 (10, 20)          |
| pairwise term    | quadratic in grid count, this does not scale indefinitely       |
| what EMR changed | the constant, not the exponent                                  |
| gated routing    | 100 / 100 / 96.6% at 3 / 4 / 5 grids, gates re-evolved per task |

**So gating fixes indiscriminate routing, not generalization.** Sparse learned inter-grid connectivity instead of all-pairs mappers is the next move, and it is not done.

# Minimal by design. And the threshold has a theory.

## WHY A TRUTH TABLE

Composition isolated from every other variable. If it wins on a task with confounds, you learn nothing. 5 frozen specialists, 1,024 rows, 30 seeds per cell.

Beyond Boolean: the continuous extension holds **100% through 8 additive gates**, so the mechanism is not gate-specific.

## WHY THE MONOLITH MUST FAIL

One CPPN must find topology, weights, *and* the spatial partitioning of activation functions, sine to the XOR region, tanh to the separable ones.

3→4 gates: that mapping search grows **combinatorially** while sub-task difficulty is unchanged (NAND is no harder than AND). Composition grows **additively**.

**Confirmed by the failure mode:** the 4-grid plateau at  $f = 0.949$  is a capacity limit, not slow search. Whether it generalizes past gate composition is, in the thesis' words, the central open question.

# I broke my own control, then rebuilt it.

## THE INVALID CONTROL · MINE

First attempt: fixed-random distractor columns, 70% success. **That control was invalid.** The values were constant per pattern, so a mapper could memorize a 16-entry lookup table instead of ignoring them.

Determinism defeated the test. So I split it into the two failure modes it had conflated.

## THE REPLACEMENTS

|             |                                                             |
|-------------|-------------------------------------------------------------|
| constant    | 47%, uninformative dims expand the search space             |
| resampled   | 100%, no lookup exists; the mapper must use the grids       |
| single grid | median 1 gen, 30/30, outputs immediately useful             |
| ablation    | grids are used; the two measures tell complementary stories |

**Closest analogy:** input dropout, implicit pressure toward grid use, not a generalization test.

# A real cousin, with three differences.

## THE DIFFERENCES

|              |                                                                                   |
|--------------|-----------------------------------------------------------------------------------|
| the features | evolved specialists on named sub-functions, not random projections                |
| the readout  | an evolved indirect-encoded network, composition is computed, not summed          |
| the evidence | an unconstrained monolith with strictly more freedom does <i>worse</i> at 4 grids |
| the reuse    | unseen compounds at 57–93%, no retraining; 100% with gated routing                |

## WHERE THE ANALOGY BITES

**Conceded:** the thesis never makes this comparison, and it should. Anyone from that literature sees it immediately.

And it bites hardest here: mappers are **re-evolved per task** , so GEENNS today is closer to frozen-features-plus-refit than to a system that generalizes. The grids accumulate; the coordination does not.

**Which is the extraction problem**, and exactly what O3 of the next program targets. See B15.

# Inspired by, not simulating. The thesis draws that line.

## WHAT IS EXTRACTED · ONE PRINCIPLE

**Architectural homogeneity with functional diversity through connectivity.** Mountcastle’s columns share a canonical microcircuit and differ by input wiring. GEENNS shares one evolved template and differs by routing.

You can see it working: HyperNEAT applies this at the *connection* level (one CPPN, all weights); GEENNS applies it at the *structural* level.

## WHAT IS REFUSED

|              |                                                               |
|--------------|---------------------------------------------------------------|
| 6 layers     | biology’s answer to axon speed, metabolic cost, volume        |
| machines     | SIMT, bandwidth, cache, so no prescribed layer count          |
| the template | a NEAT genome; evolution finds the right one for the hardware |

The honest gap: the design keeps weight plasticity in frozen grids. In the prototype, frozen means fully static.

**From prescription to emergence:** do not copy the brain’s structure, recreate the conditions under which structure emerges.

# I did. The deepest hole in the chapter.

## CONCEDED

I chose the decomposition, assigned each grid its gate, and built the compounds from them. So the prototype shows composition works **given** a good decomposition, and nothing about where one comes from.

## WHY IT IS NOT FATAL

|                |                                                                    |
|----------------|--------------------------------------------------------------------|
| the routing    | not given, mappers discovered it under evolutionary pressure       |
| Kashtan & Alon | modularly varying goals spontaneously produce modular structure    |
| Clune et al.   | connection cost produces modularity even in fixed environments     |
| the prediction | task pressure should yield reusable grids, no modularity objective |

**Structural modularity is prescribed;** whether *functional* modularity follows is left to evolution, that is O2 of the next program. I proved the mechanics of composition, not the origin of the parts.

# As an instrumented pilot. Not as a result.

## WHAT IT CLAIMS

|              |                                                                |
|--------------|----------------------------------------------------------------|
| status       | vision + proof-of-concept, never a validated result            |
| evidence     | ~3,100 runs, >40 conditions                                    |
| reporting    | confounds and limitations in the same section as findings      |
| self-flagged | among Ch 8’s eight limitations; I rank it weakest myself (Q45) |

## WHAT IT IS FOR

The published papers and the engine are the **contribution** . GEENNS is the argument for why they matter, and the evidence that the mechanics are sound rather than merely appealing.

Weighed as a reviewed result it fails, and I would agree. Weighed as a pilot that tells you which architecture survives contact with data, it does its job.

**The honest summary:** the prototype earns the next grant application, not a claim in the abstract.

■ Q30 · CH 7 “100% AT FIVE GRIDS, THEN 1 OF 29 AT SIX. IS FIVE YOUR CEILING?”

# Six is where composition inherits the failure it cured.

## THE 6-GRID COLLAPSE

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| the task        | 12 inputs, 4,096-row truth table                       |
| the floor       | all-zeros scores 0.9912, so the bar tightens to 0.995  |
| the result      | 1 of 29 seeds in 2,000 generations                     |
| the 28 failures | plateau at 0.992–0.993, true-positive rate <b>zero</b> |

## WHAT RESCUES IT

Gated routing, same six frozen specialists, same threshold: **21 of 29** at median 362 generations. A 21× improvement under identical conditions, with several distinct viable gate signatures.

So the threshold is **two-stage** : at 5 grids composition rescues the monolith; at 6, gating rescues composition.

**My headline is five deliberately** , five is where composition is 100% unaided. Six is a mitigation, not a solution, and I say so.

# At the scales I tested, no. Break-even is ~80 tasks.

## THE NEGATIVE RESULT · MINE TO REPORT

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| the sweep        | 10 pairwise AND compositions, 595 runs                        |
| success          | 94.9% augmented vs 93.3%<br>dynamic monolith: a tie           |
| cost at 10 tasks | pipeline is <b>1.34×</b> total monolithic cost                |
| break-even       | projected near <b>80</b> reused tasks (a projection, labeled) |

## WHY THAT IS THE WRONG ARGUMENT

Past the complexity threshold the comparison stops meaning anything. At 5 grids the dynamic baseline solves 0% and composition solves 100%.

When the alternative cannot do the task, 1.34× is not a tax on reuse. It is the **price of capability**.

**So the claim stays narrow:** below the threshold the monolith is competitive. I cannot claim GEENNS is cheaper, because at the scales I tested it is not.

# Not input-matched. But extra inputs cost this system.

## CONCEDED

At 5-grid the mapper sees 16 inputs (raw + 5 grid outputs); the monolith sees 11. **I ran no input-matched control.** That is a real gap in the design.

## WHY IT IS NOT THE EXPLANATION

|                     |                                                                       |
|---------------------|-----------------------------------------------------------------------|
| the distractor test | one uninformative dimension drops success 100% → <b>47%</b>           |
| so extra inputs     | expand the search space and slow convergence: a cost, not a gift      |
| single-grid control | median 1 generation: immediately useful features, not a wider space   |
| and the monolith    | has strictly <i>more</i> freedom (per-node functions) and still loses |

**If input count explained the advantage,** the constant-distractor result should have gone the other way. The grid outputs earn their place by being informative.

# One of five scales. That moved the ceiling.

## FIVE OPERATORS TESTED

|                     |                                                                       |
|---------------------|-----------------------------------------------------------------------|
| multiplicative      | 100% (2-gate) → 55% (3) → 0% (4): a near-miss: 93% at $R^2 \geq 0.80$ |
| additive mean       | <b>100% through 8 gates</b> , roughly linear cost                     |
| max · min · softmax | 0 of 29 at both 3- and 4-gate, across 174 runs                        |

## WHY IT MATTERS MORE THAN THE NUMBERS

I expected the ceiling to be **substrate capacity** , how much a frozen grid can represent. It is not. It is the **composition operator** , which is a design choice I control.

Working explanation: the smooth sin-hidden mapper is mismatched to operators with kinks at switchover. Plain optimization difficulty cannot be ruled out.

**Which is why the operator is itself a target** in the next program, rather than a settled detail.

# At the mapper, exactly. The specialists are extra.

CONCEDED

Each specialist is pop 150 × 200 generations, and at 5-grid there are five. That training is **not in the scaling table** , and the monolith never received it.

MATCHED WHERE THE CLAIM IS MADE

|                     |                                                            |
|---------------------|------------------------------------------------------------|
| mapper vs mono      | both pop 300, 500 gen                                      |
| 4-grid              | both 1,000 gen                                             |
| 5- and 6-grid       | both 2,000 gen                                             |
| the 4-grid failures | same 1,000 gen, plateau at f = 0.949–0.950 on 253/256 rows |

**A plateau is a capacity signature, not a clock signature.** It stopped improving rather than running out of time, and no budget closes 0% vs 100%. The specialist cost is measured where it belongs: 1.34× at 10 tasks (Q31).

# An OR specialist never converged. And the exclusion favors me.

## WHAT HAPPENED

|             |                                                      |
|-------------|------------------------------------------------------|
| the seed    | its OR specialist grid failed to converge, f = 0.865 |
| so          | no valid frozen specialist existed to compose        |
| the reading | excluded upstream: the composition stage never ran   |
| disclosed   | stated with its fitness in the thesis footnote       |

## AND IT CUTS AGAINST ME

That exclusion sits on **composition's** side, the condition scoring 100%. Counted as a pipeline failure instead, 5-grid composition is  $29/30 = 96.7\%$ , not 100%.

I think the upstream reading is right, because specialist convergence is a precondition rather than part of the compositional claim. The footnote gives you what you need to apply the other convention.

**What it does not change:** the monolithic baselines report 0 of 30 with a full complement. 96.7% against 0% is the same conclusion as 100% against 0%.

# Behaviorally readable. Symbolically, not at all.

## WHAT I CAN SHOW

|                  |                                                                 |
|------------------|-----------------------------------------------------------------|
| ablation         | remove grid outputs, measure the damage                         |
| weight magnitude | how strongly grid signals are read                              |
| the tension      | they do not rank the grids the same way, reported, not smoothed |
| gate signatures  | at 6 grids some seeds suppress NOR+XNOR, others OR+NAND         |

## WHAT I CANNOT

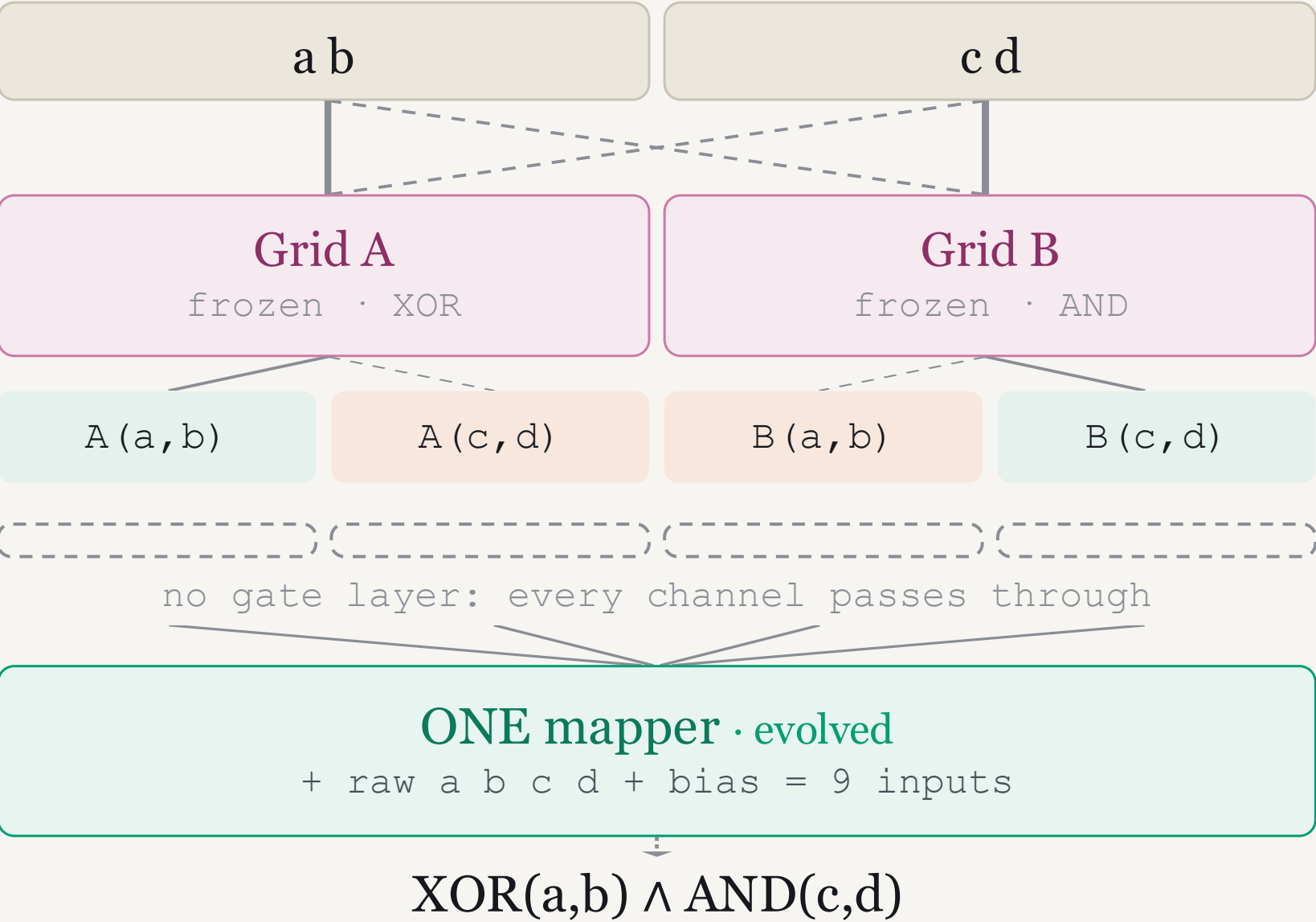
I cannot decode a mapper into a symbolic rule. There is no logical expression I can hand you and say *this is what it computes* . Interpretability was never claimed.

Routing is distributed across evolved weights in an indirect encoding, among the least legible representations available.

**The multiple gate signatures are themselves a finding:** several routings solve the same task, which is a hint about why extracting *one* general mapper is hard, there may be no canonical rule to extract.

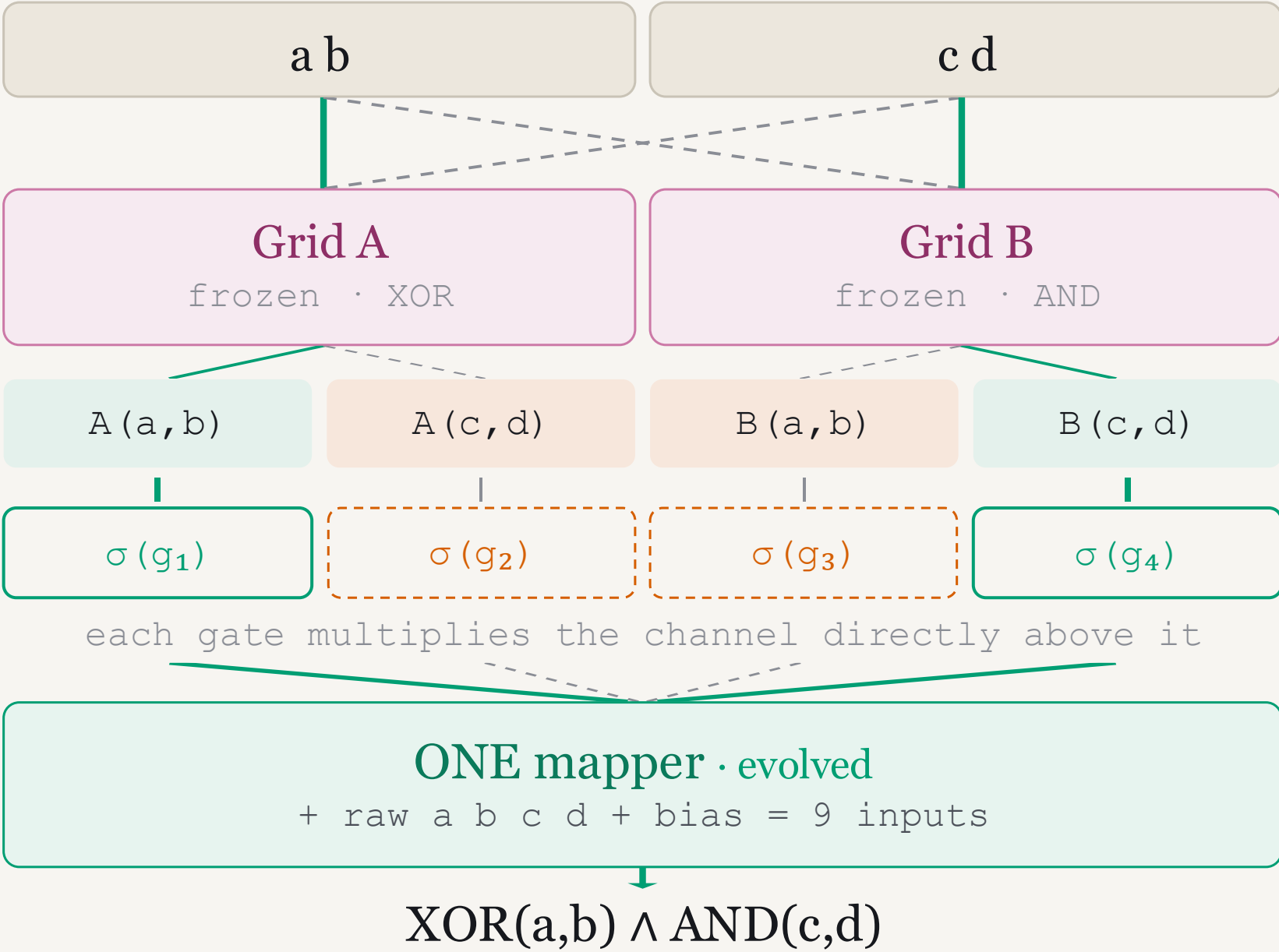
# Frozen parts, one mapper, learned combination.

## VANILLA COMPOSITION



It works, but unconstrained mappers **route indiscriminately** .

## + GATED ROUTING

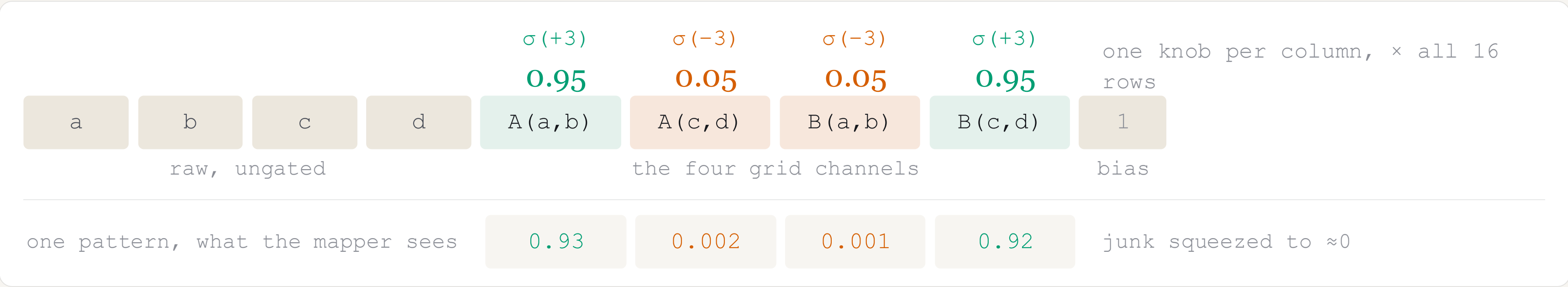


One row of knobs is the **entire** difference, unseen tasks go 57–93% → **100%**.

both pairs feed both grids · solid = grid meets its intended pair · dashed = crossed pairing

# Four volume knobs. Not a model.

Each frozen grid is wired to **both** input pairs, so 2 of the 4 channels are junk by miswiring. Without gates the mapper connects to all four about equally: it never learns to choose.



## WHY A SIGMOID

Evolution mutates an unbounded  $g$  ;  $\sigma$  makes it a legal volume. **Attenuate-only** (never amplify or flip), and  $\sigma(0)=0.5$ : every knob starts half-open, so selectivity must be *discovered*.

## WHERE THE GATES LIVE

**Not in the NEAT genome.** Each individual carries a private 4-vector. NEAT evolves networks; knobs follow copy-the-winners (top 10% kept, Gaussian noise on the rest). Coupled only by the *shared fitness*.

# It isn’t sampled, and there are no distractor grids.

## THREE THINGS THAT ARE NOT HAPPENING

|                       |                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| no special grids      | both specialists trained the ordinary way on their own 4-row table, then frozen                                 |
| no sampling           | the <b>complete</b> compound truth table every generation, every individual, 16 rows at 2 gates, 4,096 at 6     |
| no labels at run time | correct/distractor are measured <b>after</b> the run. The only signal evolution ever sees is the compound score |

A *correct* channel is a grid fed its intended pair; a *distractor* is the same grid fed the other pair. **Junk by miswiring, not by nature.**

## AND WHAT EVOLUTION DID WITH IT

|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| selective | 8.28× median gate ratio , junk muted, exactly as hoped                         |
| inverted  | 0.63×, solved it with the knobs <i>backwards</i> , recruiting junk as features |
| neutral   | 1.05×, both grids compute the same signal, so the knobs did not matter         |

All three reach **100%** (57–93% without). So the knobs make selection *possible* , evolution uses it only when the task rewards it.

Note: the noise-control experiments use “distractor” for random input dimensions, unrelated to gating.

# A second population, beside NEAT, not inside it.

Every individual carries a genome *and* a private 4-vector of knobs. NEAT evolves the genomes. The knobs run their own small evolution strategy.

## 1 INITIALIZE

A 4-vector near zero for every slot, so  $\sigma(\mathbf{g}) \approx 0.5$ , every channel half-open. Selectivity must be discovered.

```
randn(pop, 4) × 0.1
```

## 2 SCORE AS A PAIR

The knobs gate that individual's own channels, then its network runs. **Never scored apart**: a knob is judged only by the mapper using it.

```
fitness = 1 - MSE
```

## 3 RANK & KEEP

Rank everyone by that fitness. The **top 10%** of knob vectors survive unchanged as elites.

```
elites = top 10%
```

## 4 REPOPULATE

Every slot takes a parent from the elite pool, cycled through it. Winning routings spread across the population.

```
parent = elites[k mod n]
```

## 5 MUTATE

Gaussian noise on every non-elite copy: small changes to  $\mathbf{g}$  are small changes in routing.

```
+ N(0, 0.3)
```

## HOW THE TWO POPULATIONS COUPLE

NEAT never sees the gates. The script computes the gated fitnesses and **replays NEAT's selection** with them, so both populations are judged by identical scores. No genes are exchanged: the *shared fitness vector* is the only channel between them.

## AND THE LIMIT

This loop is **per task**. Nothing carries a routing solution to the next one.

# The truth table is four corners of a surface.

Each gate becomes the smooth polynomial that **matches its truth table at the corners** of  $[0,1]^2$ .

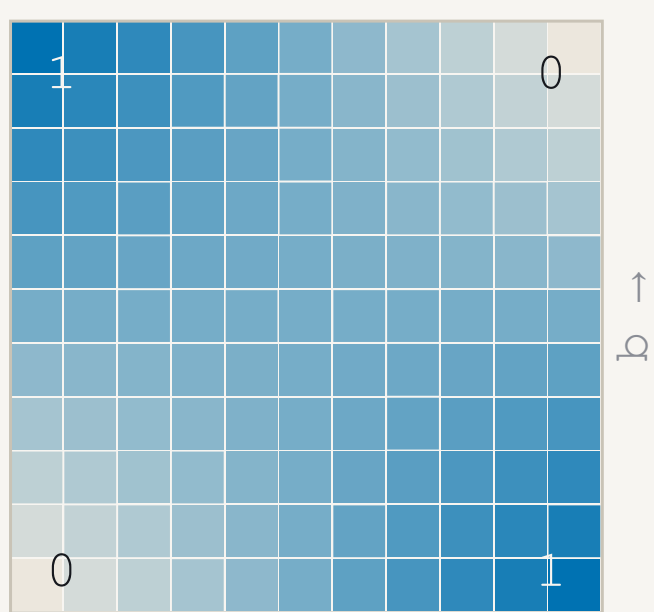
BOOLEAN XOR

| a | b | out |
|---|---|-----|
| 0 | 0 | 0   |
| 0 | 1 | 1   |
| 1 | 0 | 1   |
| 1 | 1 | 0   |

4 rows · discrete

→  
interpolate

CONTINUOUS XOR<sub>c</sub>



a saddle, corners match the table,  
center is 0.5

THE POLYNOMIALS

|                   |               |
|-------------------|---------------|
| XOR <sub>c</sub>  | $a + b - 2ab$ |
| AND <sub>c</sub>  | $c \cdot d$   |
| OR <sub>c</sub>   | $e + f - ef$  |
| NAND <sub>c</sub> | $1 - gh$      |
| NOR <sub>c</sub>  | $(1-i)(1-j)$  |

Plug in 0s and 1s → **exactly** the Boolean answer. (+ XNOR, IMPLY, NIMPLY for 6–8 gates.)

## THE SPECIALISTS ARE ORDINARY REGRESSORS

Trained exactly like the Boolean ones, but the target is the surface: regression on an **8×8 grid** of  $[0,1]^2$ , sin hidden, threshold 0.93, then frozen.

## AND THE SCORE CHANGED

A product of numbers  $\leq 1$  is near zero almost everywhere, so under 1–MSE “predict the mean” scores well. **R<sup>2</sup> divides by the target’s variance**, predicting the mean scores exactly 0. Bar:  $R^2 \geq 0.85$ .

# The mapper doesn't apply it. It *defines the target*.

## WHAT THE MAPPER IS TOLD

|                  |                                                                                                |
|------------------|------------------------------------------------------------------------------------------------|
| before evolution | the script applies the operator to the <i>ideal</i> formulas at every sample → a target column |
| the mapper sees  | only the frozen outputs and a bias, in composition-only, <b>no raw inputs at all</b>           |
| the rule         | <b>never given.</b> It must evolve any function that matches the target                        |

**Worked sample.** (a,b)=(0.5, 0.5), (c,d)=(0.8, 0.9) → XOR<sub>c</sub>=0.5, AND<sub>c</sub>=0.72, target **0.36** . The mapper receives [0.5, 0.72, 1] and must output 0.36, at the corners, the Boolean case exactly.

## WHY PRODUCTS FAIL

|                    |                                                                                        |
|--------------------|----------------------------------------------------------------------------------------|
| the mechanism      | each specialist is good but imperfect, and sigmoid <b>compresses toward the middle</b> |
| multiply k of them | the compressions <b>compound</b> : every extra gate is worse                           |
| average them       | the errors <b>cancel</b> , more gates makes the mean more stable                       |

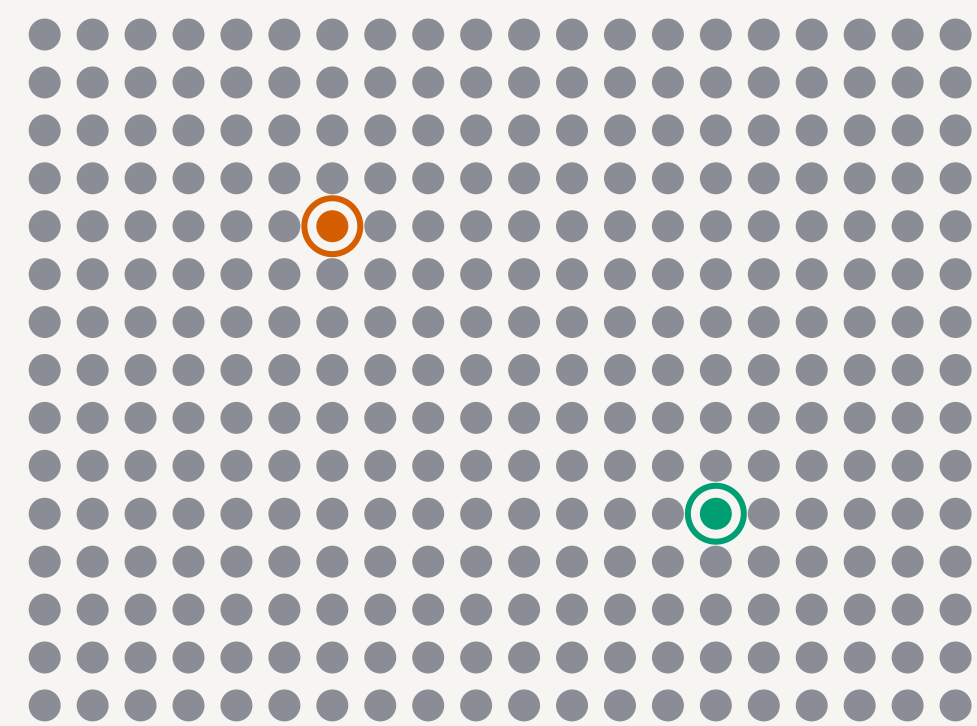
## AND THE MAPPERS KNOW IT

Literal multiplication caps at  $\prod R^2_i \approx$  **0.764** at 4-gate. **All 29 seeds beat it.** Probing shows a degree-3 polynomial (96.5% of variance), not a rescaled product: it refuses to compound compression.

Conjunction hard-wires multiply-like composition, so in the Boolean regime the operator was **invisible** . Continuous targets make it a free axis, and the bottleneck: **not the grids, not the mappers, not the budget.**

# The solver exists. Selection can't see it.

ONE SOURCE-EVOLVED POPULATION  
· 300



- the **champion** scores 0.53–0.80 on the unseen target → fail
- a **latent solver**  $\geq 0.95$  with no retraining, and *not* the champion

THE CENSUS · 29 SEEDS, ALL 300 EVALUATED

|                |                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------|
| latent solvers | 1.03 and 1.21 per 300, 0.345% / 0.402%                                                               |
| coverage       | 28 of 29 seeds contain at least one                                                                  |
| why invisible  | source fitness never rewards generality, so the generalizing individual is unremarkable <i>there</i> |

IS 1-IN-300 JUST LUCK? THE RANDOM CONTROL

|                       |           |                        |
|-----------------------|-----------|------------------------|
| 0                     | 0.0017%   | 0.0006%                |
| pop 300 · 58/58 evals | pop 1,000 | pop 3,000 · 10× budget |

A ~600× density gap. Source evolution moved the population somewhere sampling cannot reach.

# It refines the question. It doesn’t reverse the answer.

**1 gen** Warm-start’s median on **all four** targets. So 45–79× and 223–351× are the cold medians divided by one: it doesn’t search faster, it **starts where a solver already is**.

## WHAT STANDS

|                    |                                    |
|--------------------|------------------------------------|
| individual mappers | still do not zero-shot             |
| the champion       | still fails, 0.53–0.80             |
| the test’s verdict | untouched: I am not withdrawing it |

## WHAT FALLS

|                    |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| one inference only | “champion fails” ⇒ “capability absent”                                                |
| Jaccard            | median <b>1.0</b> over 87 pairs: <i>one</i> individual solves source and both targets |
| so the question    | “does it generalize?” → “how do we <b>extract</b> the 0.3%?”                          |

**AND IT IS CONDITIONAL**  
Clean under *recomposition* of the same parts. Substitute a gate (NOR→XNOR) and it thins to 2.06×, with only 4/29 holding a zero-shot solver. Partly compositional, partly **distributional**.

# Chapter 8 ships the table. Here it is.

|                                |                                                       |
|--------------------------------|-------------------------------------------------------|
| MNIST 43% · ch4                | deep learning >99%; the contribution is the diagnosis |
| Two Spirals 0% · ch6           | architecture-limited; neither ES- nor EMR solves it   |
| XOR-led speedups · ch6         | 15 problems mitigate it; industrial scale untested    |
| No gradient baseline · all     | evolutionary only, by argued design                   |
| SIMT-scoped proof · ch5        | dataflow and FPGA untested                            |
| Single-objective · ch3-6       | no Pareto, no multi-objective                         |
| No zero-shot composition · ch7 | fitness 0.53–0.80 against a 0.95 bar                  |
| CPU-only · ch3-4               | EMR did not exist yet when they ran                   |

**If forced to pick two:** GEENNS carries the most narrative weight on the least-reviewed evidence; and partitioned experts was validated on one configuration, one dataset. What I would do differently: build the engine first.